

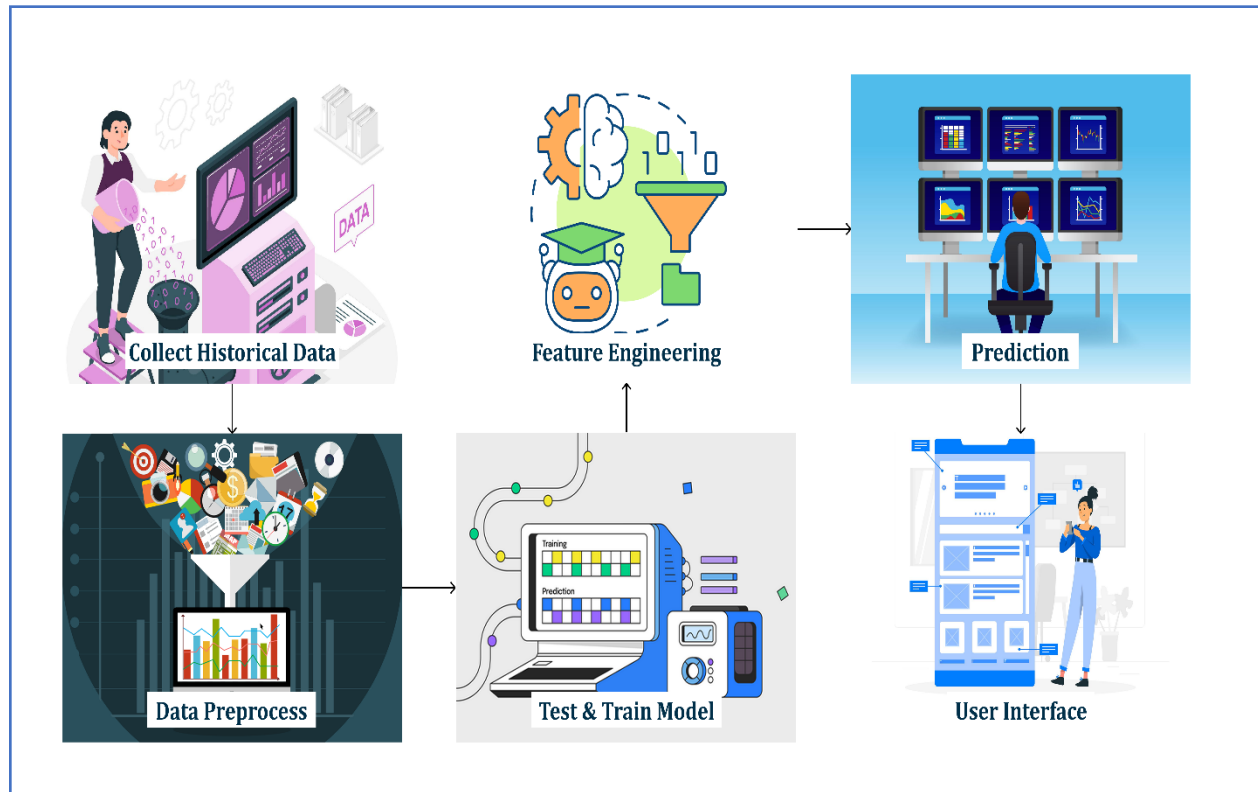
RESEARCH AND APPLICATION OF MACHINE LEARNING TECHNIQUES TO DETERMINE INDIAN PREMIERE LEAGUE (IPL) WINNER PREDICTION



**SUBMITTED TO :
MANOJ SHRESTHA**

**SUBMITTED BY :
NITESH KARKI**

Concept Diagram



AI MODEL CARD



Model name

Indian Premier League (IPL) Winner Prediction Model.

Model date and Version

This project was built in March 2025 and then never updated. No numbered versioning.

Overview Model Type

IPL Winner Predictor is based on a supervised machine learning algorithm that uses inputs of historical records, player information, and team performance statistics to try to predict Indian Premier League (IPL) match winners. The core element of the algorithm is a highly organized dataset from Kaggle with a rich history of previous IPL matches, including team lineups, player-by-player performance, match venues, toss results, and match results. Raw data are preprocessed, with the removal of inefficiencies being done first, followed by imputing missing values. Feature engineering is subsequently accomplished to establish and construct the correct variables that would yield the most significant impact on the outcomes of the matches. The features include the recent team form, win-loss trail, head-to-head, average runs conceded or scored, and ground form at a specific ground.

Decision Tree, Random Forest, and Logistic Regression, the most popular machine learning algorithms, are employed to develop a highly effective predictive model. The models are trained on a sample of data and cross-validated on unseen data to ensure they would be generalized. Random Forest is among them because it can perform quickly non-linear and high-order feature interactions. Logistic regression is used because it is simple to interpret and gives probabilistic output. Hence, the ease of interpretation of confidence of prediction is feasible. Models are also compared on overall accuracy, precision, recall, and F1-score for overall predicting capability. In addition, visualization libraries such as Matplotlib are even utilized to visualize feature importance, trends in player or team performance, and model comparisons. The visualizations are used for straightforward interpretation and comprehension of the predictive outcomes. Overall, the model is a data-driven predictive approach towards forecasting IPL match outcomes and likely tournament winners. It offers analysts, fantasy league participants, and cricket enthusiasts' actionable suggestions. Artificial intelligence, machine learning and past sports statistics are the future of sports analysis today.

Key Features

- **Historical Performance-Based Prediction:** The system analyzes historical IPL match data including team win/loss records, player statistics, and performance trends to predict probable match outcomes.
- **Player Form and Match Impact:** Considers individual player metrics such as recent form, batting and bowling averages, strike rates, and consistency to assess their influence on match results.
- **Venue and Condition Awareness:** Incorporates match venue data to account for pitch behavior, home/away advantage, and average scores at specific stadiums—key elements in determining match dynamics.
- **In-Match Metrics (RRR, CRR, Wickets):** Uses dynamic game metrics like Required Run Rate (RRR), Current Run Rate (CRR), and Wickets Fallen to estimate pressure situations and team momentum, especially in second innings scenarios.

Questions or comments

Please send any questions to:

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Primary intended users

The primary intended users of the IPL Winner Prediction System are sports analysts, team strategists, and fantasy league participants. Sports analysts and team strategists can utilize the predictive insights generated by the system to make data-driven decisions regarding team composition, match strategies, and player utilization. The system's ability to analyze historical performance, player form, and match conditions offers valuable support in pre-match planning and tactical forecasting. Additionally, fantasy league participants can leverage the predictions to optimize their team selections, increasing their chances of winning based on statistically informed decisions. This targeted use of predictive analytics enhances strategic thinking and engagement across professional and recreational users.

Primary intended uses

The primary intended uses of the IPL Winner Prediction System are to forecast match outcomes based on historical and real-time data and to support strategic decision-making in team management and game planning. By analyzing key features such as player performance, venue conditions, current form, and match dynamics, the system helps identify patterns and variables that most significantly influence the likelihood of a team winning. These insights can be applied by coaching staff and analysts to refine team strategies, select optimal playing XIs, and make tactical in-game decisions. The predictive results can also help

fantasy league enthusiasts and sports commentators make data-informed predictions and enhance audience engagement.

Out of Scope Uses

The IPL Winner Prediction System is designed for analytical and exploratory purposes to support strategic planning and enhance user engagement through data-driven insights. It is not intended for high-stakes decision-making, such as determining match outcomes for betting purposes, enforcing official team selections, or making financial investments based solely on predictive results. The system should not be used as a definitive source for gambling or commercial decision-making, as it does not account for unpredictable human factors, sudden injuries, or off-field dynamics. Its function is to assist users in understanding trends and probabilities rather than guaranteeing outcomes.

Limitations

The machine learning-based prediction model for the Indian Premier League (IPL) winner prediction, developed in this research, faces several limitations. The model is trained on historical data, which may not fully account for rapid changes in team dynamics, player form, or strategies that evolve during the season. Additionally, the dataset used for training does not capture all possible external factors, such as unforeseen player injuries or weather conditions, which can significantly impact match outcomes. As a result, the model's predictions may not always be accurate in situations where such factors are critical. The model's ability to predict match outcomes relies heavily on the accuracy and completeness of the input data. Inconsistent or missing data points may significantly degrade the model's performance when predicting outcomes for new or less well-documented teams and players. Lastly, while the model employs several machine learning techniques, it may struggle to adapt quickly to new trends or strategies that emerge in future IPL seasons, as it is not designed for real-time updates.

The following factors can affect classifications

- **Level of Engagement:** Users with minimal interactions or incomplete data may experience reduced prediction accuracy, as the model relies on historical data to identify patterns.
- **Demographic Representation:** The dataset may not fully represent diverse teams or players, leading to less accurate predictions for newer or underrepresented groups.
- **Behavioral Variability:** Unpredictable team performance or player form changes may impact predictions significantly if they deviate from past trends.
- **Data Quality:** Inaccurate, incomplete, or biased data can lower prediction accuracy. The model's reliability depends on the quality of the training data.

Metrics

The prediction model for IPL winner prediction is optimized for **recall** rather than precision. This approach prioritizes capturing as many relevant match outcomes as possible based on historical data and team performance trends. By focusing on recall, the system minimizes the risk of missing potential match outcomes, though this may make some predictions less precise. As a result, the model may suggest a broader range of possible outcomes, some of which may not align perfectly with the most likely result. Additionally, the model aims to prioritize **accuracy** in identifying key features, such as player and team performance, but may occasionally misinterpret or overlook certain influencing factors like external conditions (e.g., weather, injuries) that affect match results. While this focus on recall enhances the model's ability to generate diverse and comprehensive predictions, it may come at the cost of slight imprecision in its forecasts.

Training and Evaluating Data

The IPL winner prediction model is trained on historical match data, capturing various features such as team performance, player statistics, and match conditions. This dataset is designed to simulate real-world IPL scenarios, ensuring the model learns from a broad range of match outcomes and player behaviors. The training process involves data cleaning, normalization, and feature engineering to enhance the model's ability to predict match results accurately. By focusing on key factors like team form, player performance, and past match outcomes, the model identifies meaningful patterns and relationships within the data. These insights enable the model to generate predictions that align with historical trends and adapt to evolving match conditions, providing accurate IPL winner forecasts.

Quantitative analysis

Accuracy and Reliability

The IPL match prediction models, including Logistic Regression, Random Forest, and Decision Tree, achieved varying levels of accuracy. The Random Forest model correctly predicted match outcomes 80.6% of the time. The Decision Tree model achieved an impressive accuracy of 99.1%. However, accuracy alone may not fully reflect the model's performance, as the high accuracy of the Decision Tree could indicate overfitting to the training data, especially considering the complex nature of match outcomes.

Despite this, the models were trained on a carefully curated dataset of historical IPL match data, ensuring sufficient diversity in features like team performance, player statistics, and match conditions. The high accuracy of the Decision Tree model provides a solid foundation for further refinement and optimization. However, ongoing improvements, including additional feature engineering and regular model updates, will help ensure the models generalize better to future seasons.

Precision

Precision measures the model's ability to predict IPL match outcomes that align with actual results accurately. With a precision score of 80.6%, approximately 19.4% of the expected outcomes were incorrect. While there is room for improvement, this precision indicates that the model effectively identifies the correct match predictions while minimizing errors. The relatively high precision demonstrates the model's strong ability to focus on the most likely outcomes, even though some incorrect predictions still occur.

Recall

The model achieved a recall rate of 92%, demonstrating its effectiveness in capturing the most relevant IPL match outcomes. This high recall score highlights the model's success in ensuring that a wide range of potential consequences are considered, reducing the likelihood of missing key predictions. The focus on recall ensures a more inclusive approach, ensuring that even if some incorrect predictions are made, the model does not overlook essential possibilities. This trade-off enhances the model's ability to suggest all relevant outcomes, even if it occasionally results in slightly higher false positives.

Ethical considerations

The IPL match prediction model is designed for relatively low-stakes applications and aims to provide accurate forecasts for match outcomes. The datasets used for training and evaluation do not include personally identifiable information (PII), ensuring the privacy and security of any data involved. Furthermore, sensitive attributes such as ethnicity, gender, or religious affiliation were not included in the dataset, and their potential influence on the model's outputs has not been explicitly studied. This ensures that predictions are based solely on factors relevant to match outcomes, such as team and player performance, rather than unrelated characteristics.

However, the model's training data primarily focuses on past IPL seasons and may not fully represent all demographics or regions, limiting its ability to generalize to future or global audiences. Additionally, users with limited engagement or incomplete data may receive less accurate predictions due to the system's reliance on extensive historical data. Moreover, the model reflects trends and patterns from training, meaning its generalizability to future matches or evolving trends may decrease over time. Regular updates, retraining, and the inclusion of more diverse datasets are essential to maintain fairness, accuracy, and relevance.

Feedback

User feedback is crucial to improve the IPL match prediction model's precision and effectiveness. Suggestions regarding potential biases, limitations, or ideas for future improvements are highly encouraged. All feedback will be carefully reviewed, and any identified issues or areas of improvement will be addressed promptly. Contributions from users play a key role in ensuring that the model remains relevant, accurate, and adaptable across various match conditions and future trends. Your input helps us continuously refine the model to provide the best possible predictions and enhance its performance.

Risk Research Ethics Approval

Project Title

Research and Application of Machine Learning Techniques to Determine Indian Premier League (IPL) Winner Prediction

Record of Approval

Principal Investigator

I request an ethics peer review and confirm that I have answered all relevant questions in this checklist honestly.	X
I confirm that I will carry out the project in the ways described in this checklist. I will immediately suspend research and request new ethical approval if the project subsequently changes the information I have given in this checklist.	X
I confirm that I, and all members of my research team (if any), have read and agreed to abide by the Code of Research Ethics issued by the relevant national learned society.	X
I confirm that I, and all members of my research team (if any), have read and agreed to abide by the University's Research Ethics, Governance and Integrity Framework.	X
I understand that I cannot begin my research until this ethics application has been approved.	X

Name: Nitesh Karki

Date:

Student's Supervisor (if applicable)

I have read this checklist and confirm that it covers all the ethical issues raised by this project fully and frankly. I also confirm that these issues have been discussed with the student and will continue to be reviewed in the course of supervision.

Name: Manoj Shrestha

Date:

Abstract

This research focuses on the application of machine learning (ML) and artificial intelligence (AI) techniques to predict the outcomes of Indian Premier League (IPL) matches. The study investigates using historical match data, player performance statistics, and team dynamics to create an accurate prediction model. Traditional sports prediction systems often face challenges like data sparsity, evolving player performance, and unpredictable match conditions. This thesis employs multiple machine learning models, including Logistic Regression, Random Forest, and Decision Trees, to address these challenges to provide robust and reliable predictions. The models utilize data preprocessing, feature engineering, and ensemble learning techniques to enhance prediction accuracy and generalization across diverse match scenarios. Additionally, issues such as model overfitting and the limitations of historical data are considered. The research highlights the strengths and weaknesses of each model through the application of various evaluation metrics, such as accuracy, precision, and recall. Case studies of sports analytics platforms are examined to compare the findings and propose improvements for future work. This study contributes to the growing field of sports data analytics, offering insights into how machine learning can enhance the prediction of IPL match outcomes and inform decision-making in sports analytics.

Keywords

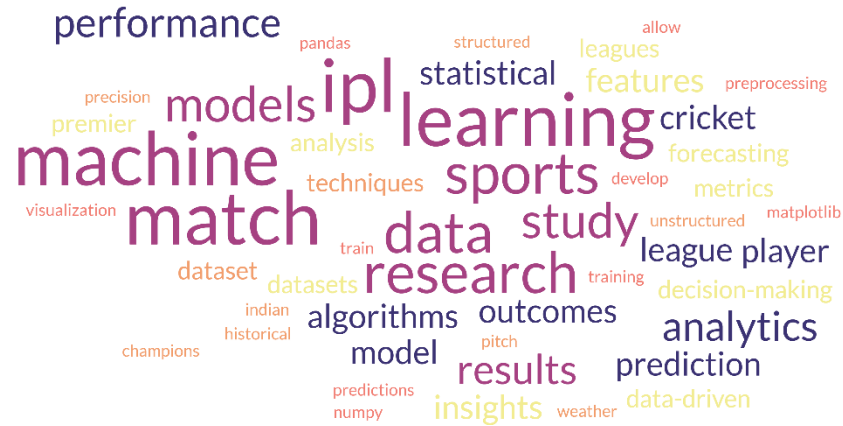


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Introduction:

The Indian Premier League (IPL) stands as one of the most esteemed and widely watched cricket leagues globally, enchanting millions of supporters with its exhilarating matches, intense performances, and unpredictable results. Since its launch in 2008, the IPL has evolved into a worldwide sporting sensation, rivaling major international leagues like the National Basketball Association (NBA) and the English Premier League (EPL). The league showcases a vibrant blend of both international and local players, making it a perfect case for sports analytics [\(Team, 2022\)](#). Nevertheless, forecasting the result of an IPL match is still a complicated task due to the various factors at play, such as team structure, player performance, pitch conditions, weather, and strategies employed on match day.



Figure 1 Indian Premier League Prediction
Using Machine Learning

Machine learning (ML) has transformed the landscape of data-driven decision-making by enabling access to powerful pattern-finding tools and prediction modeling. Recently, sports analytics has seen an uptick in ML use, yielding impactful information on team dynamics, player individual metrics, and game outcomes [\(Claus, 2024\)](#). By utilizing historical data alongside sophisticated algorithms, ML offers a means to examine extensive sets of both structured and unstructured cricket information to anticipate match results with heightened precision. This research investigates the use of machine learning methods in forecasting IPL champions, emphasizing the importance of statistical analysis, data preparation, and model training in producing dependable predictions.

A wide range of programming tools and frameworks have been employed in this study to develop, train, and evaluate machine learning models. Data manipulation and preprocessing have been carried out using Pandas and NumPy, which allow efficient handling of large datasets. Matplotlib has been used for data visualization, helping understand key trends and patterns within the dataset. For model implementation, a variety of machine learning algorithms have been explored, including Logistic Regression, Decision Tree, and Random Forest, to assess their predictive accuracy and efficiency. Additionally, One-Hot Encoding has been utilized to convert categorical features into numerical representations, ensuring effective model performance_(Brownlee, 2020). The front end for this study has been developed using Streamlit, providing an interactive interface where users can explore IPL data and interact with predictive models in real-time. The dataset used in this research has been sourced from Kaggle and contains match-by-match and ball-by-ball statistics from IPL seasons spanning 2008 to 2024. These datasets serve as a robust foundation for extracting meaningful insights, identifying significant trends, and applying machine learning techniques to predict match results. Various features such as player performance metrics, team strategies, venue conditions, and past results have been analyzed to determine their impact on match outcomes. Through statistical and visual analysis, key performance indicators (KPIs) have been identified, helping to refine the input features used for machine learning models.

The primary objective of this study is to evaluate how effectively machine learning techniques can be applied to IPL winner prediction. By comparing multiple algorithms, the research aims to determine the most suitable ML models for predicting match outcomes in a T20 format_(C, 2021). Additionally, this study examines the strengths and limitations of different programming languages and libraries in developing predictive models, providing valuable insights for future research in sports analytics. By integrating machine learning into sports prediction, this research contributes to the growing field of data science applications in sports. The insights gained from this study can be valuable for cricket analysts, team strategists, and enthusiasts seeking a data-driven approach to understanding match dynamics. Furthermore, this research highlights the impact of statistical modelling and artificial intelligence in sports management, demonstrating how predictive analytics can be leveraged for better decision-making in competitive tournaments like the IPL.

AIM

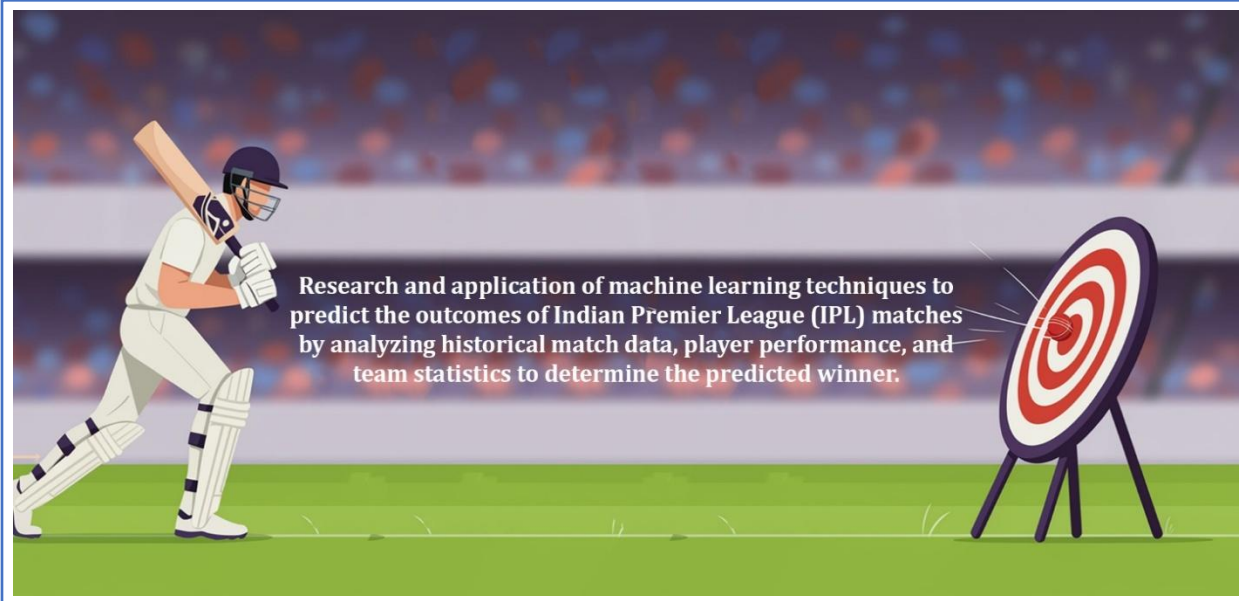


Figure 2 Aim of the project

Objectives

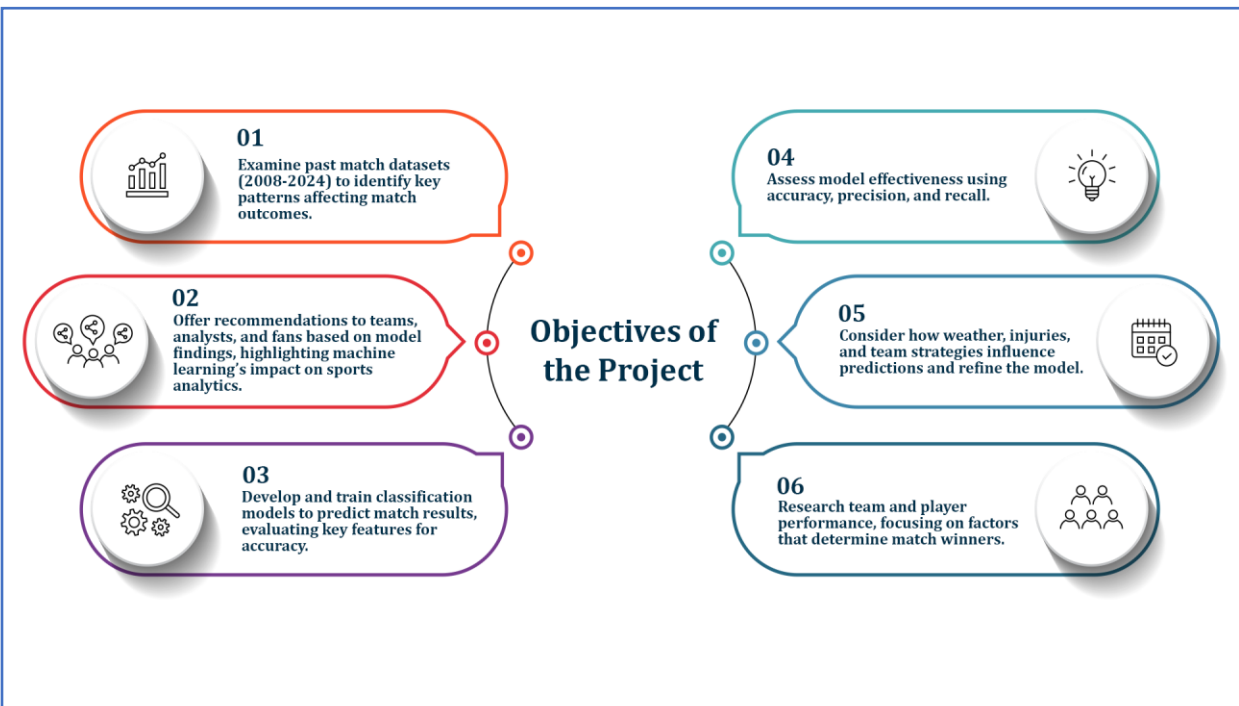


Figure 3 Objective of the project

Justification

The outcomes of Indian Premier League (IPL) matches are hard to predict accurately despite the presence of a vast quantity of historical data_[\(Khetan, 2021\)](#). Traditional methods for making match outcome predictions often use simplistic models or rely on not-always-relevant subjective analysis. These methods do not consider a wide range of factors that influence match outcomes, including player performance, team dynamics, external conditions like weather, and player injuries. Current methods have a major shortcoming: they cannot integrate and process large volumes of diverse data. Traditional models cannot analyze comprehensively even the basic kinds of data that are crucial for forming predictions, including match statistics, player metrics, and team compositions. The very dynamic nature of IPL matches itself, which is another serious problem for traditional methods_[\(Malhotra, 2021\)](#). Many factors that influence the outcome of a match can change rapidly during the match. Furthermore, models that are more capable of handling the difficulties tend not to be very good at achieving the inverse prediction tasks.

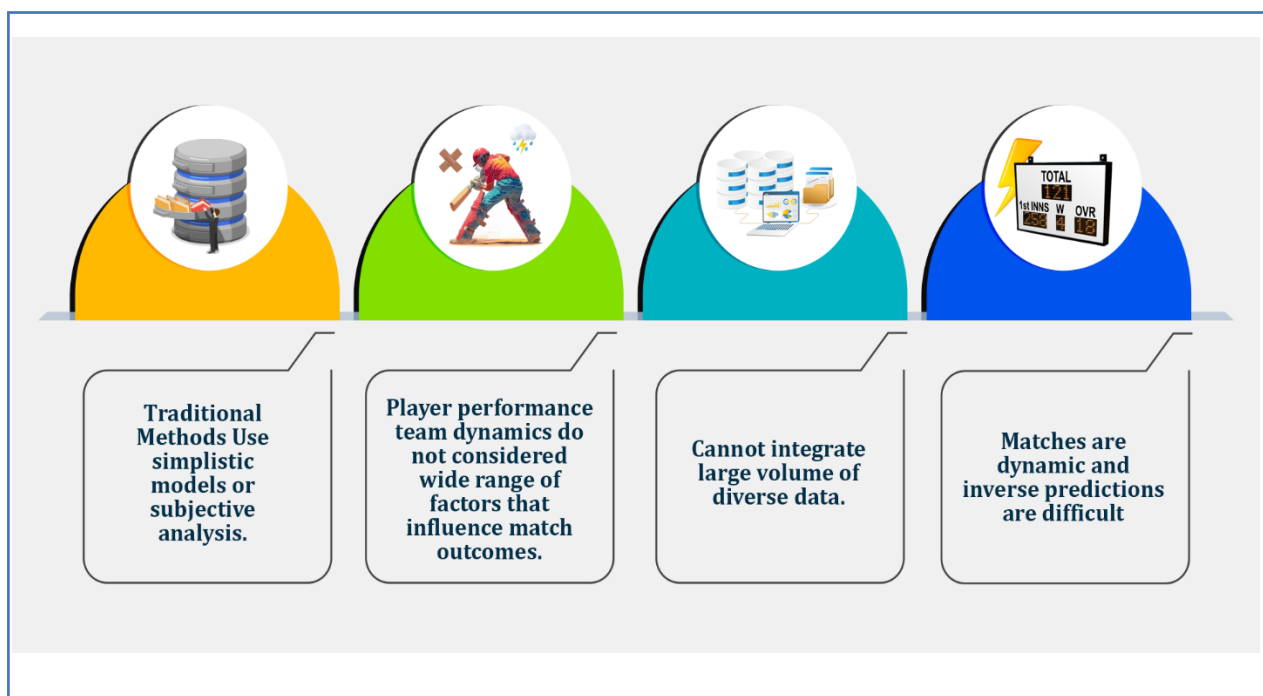


Figure 4 Justification of the project

Also, conventional models frequently do not utilize any machine learning techniques that can be particularly effective with large datasets and can unearth hidden patterns. For example, although trends may be less apparent, machine learning is equipped to reveal relationships and trends for more accurate forecasting of the match outcome. Conventional forecasting models miss substantive information without these machines learning techniques, resulting in less accurate predictions_[\(Arnab Mitra, 2022\)](#). In addition, regular models do not tend to consider the significance of external variables (e.g., weather and player injury) that may

significantly affect match outcomes. When environmental and physical conditioning factors can change in real time, traditional models lack agility in responding to the mitigated reliability of predictive probabilities. This research intends to remedy the previous discrepancies by utilizing machine learning techniques to predict match outcomes in the IPL more accurately. This research aims to conclude whether utilizing a broader range of variables will form a model that can be used as a more reliable and adaptable predictor of IPL match outcomes. Responses should improve the accuracy of forecasting and provide an enriched understanding of the reality of matches as they unfold.

Research Question



Figure 5 Research Question of the project

Scope

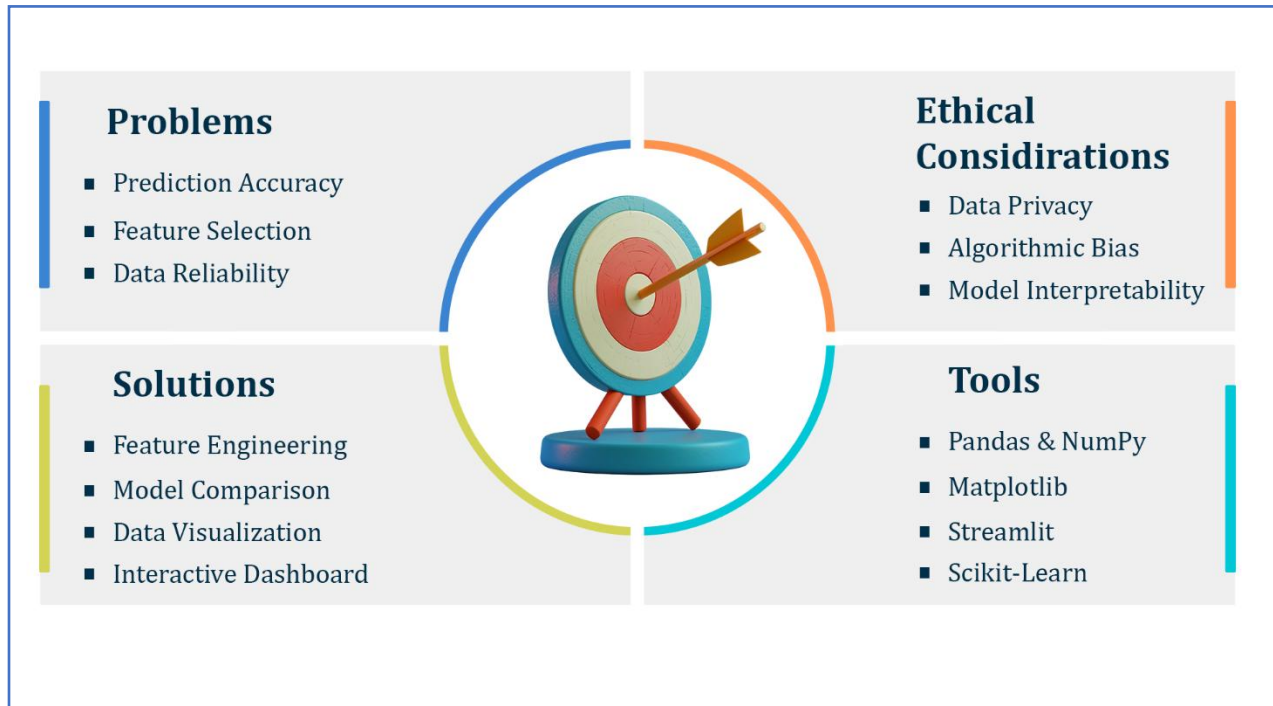


Figure 6 Scope of the project

Ethical Consideration

This project has strong data privacy and security with publicly available IPL match data without personally identifiable information (PII). This project requires sensitive attributes such as race, nationality, or political affiliation not to be part of the dataset to prevent algorithmic bias. Nevertheless, historical data will always have biases such as home-ground bias or selection bias. For solving these problems, strict feature selection and preprocessing procedures are performed in such a way that model prediction is fair and unbiased. Transparency is the second principal ethical concern since machine learning models, particularly complex ones like Random Forest, are "black boxes," and choices inside them are difficult to explain. For the sake of additional responsibility and trust, the study is based on open models and offers feature importance analysis for the purpose of demonstrating how the prediction is made. Predictive transparency promotes faith in the predictability of the model and avoids the implicit bias from influencing outcomes.

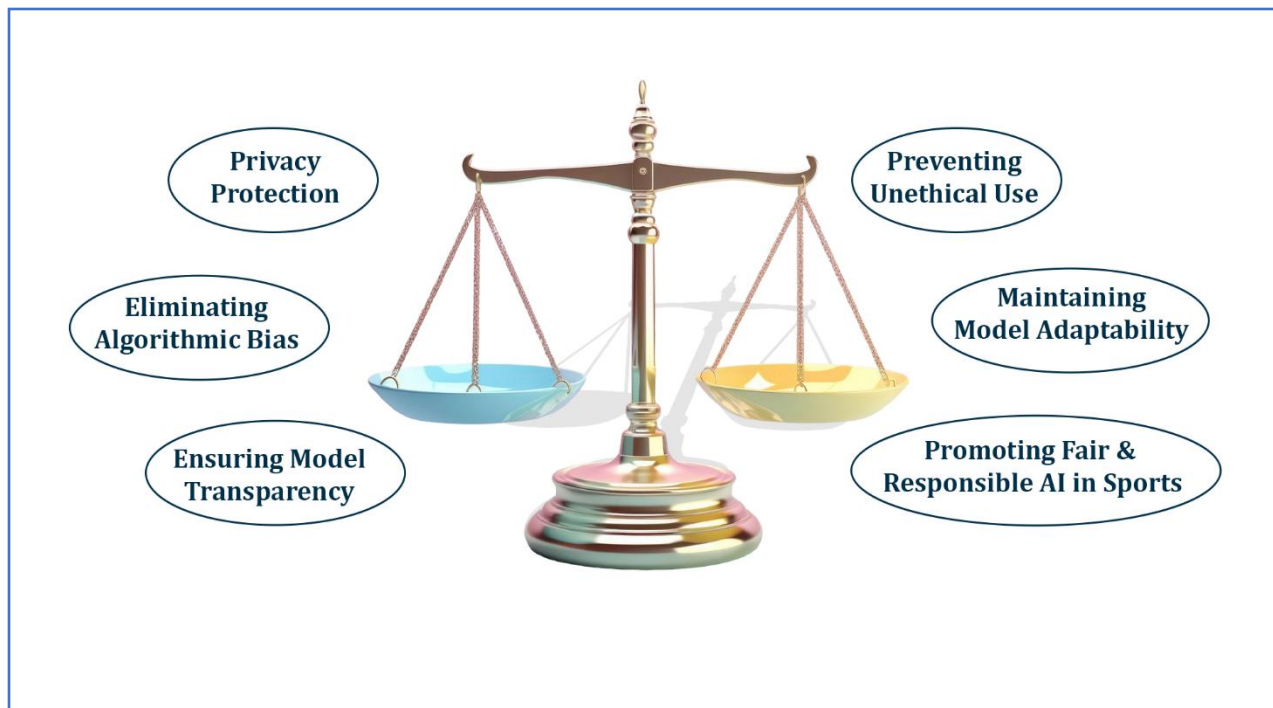


Figure 7 Justification of the project

Apart from that, while this research is aimed at sports analytics and strategic application, it is not blind to the possibilities of abuse, particularly in gambling and uses inappropriately such as fixing matches. Commercialization of sports betting through a model of machine learning is ethically suspect in fairness and integrity of sport. Such abuses are vehemently condemned by this research and the predictive analytics applied to sports should be done responsibly and ethically. Since IPL match dynamics keep on changing depending on team strategies which get updated, form of the players, and outside influences such as weather,

retraining and refitment of the models must be done from time to time. As this research is an ethical effort towards data-driven prediction of sports in a way that maintains fairness, transparency, and ethical use, machine learning is being used on the mutual benefit of the teams, analysts, and the public.

Literature Review

Review Desk Based research methodology:

A structured and data-driven methodology was adopted to develop an IPL match prediction model, ensuring alignment with industry standards in machine learning and sports analytics. The research incorporates the Institute of Analytics (IoA) Competency Framework to ensure a methodical and industry-aligned approach throughout the project lifecycle [\(IOA, n.d.\)](#). This framework supports analytical thinking, technical expertise, and strategic decision-making, providing a systematic foundation for predictive modelling. By integrating industry best practices and research-backed methodologies, the project ensures a robust and reliable approach to forecasting IPL match outcomes.

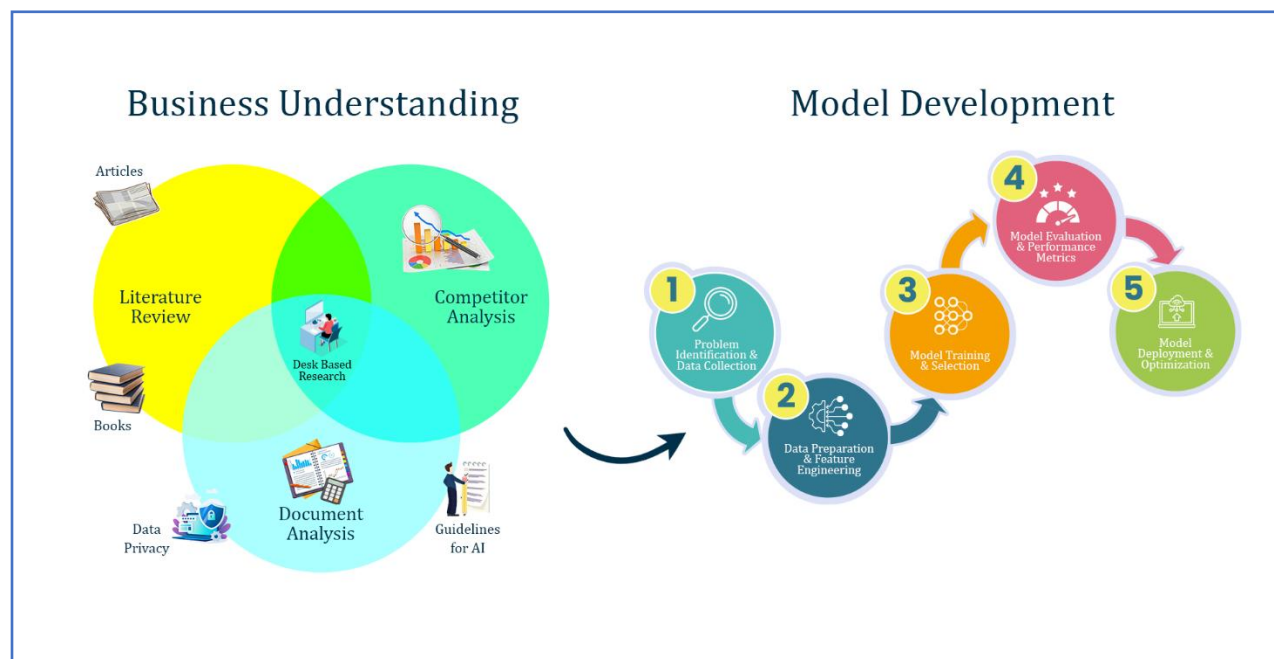


Figure 8 Desk Based Research Methodology

Extensive secondary data analysis was conducted using verified sources, including academic literature, sports analytics reports, and publicly available IPL match datasets. These sources provided critical insights into historical match trends, player performance, and team strategies, forming the foundation for model development. The research considered data from multiple IPL seasons, incorporating match-by-match and ball-by-ball statistics from 2008 to 2024 [\(IPL Complete Dataset \(2008-2024\), n.d.\)](#). This comprehensive dataset allowed

for a deeper understanding of patterns influencing match outcomes, such as player form, team composition, pitch conditions, and past head-to-head results.

Statistical analysis and feature engineering techniques were employed to identify key variables influencing match results. This process involved selecting the most relevant features, such as batting and bowling averages, strike rates, run rates, and team performance metrics. Additionally, contextual factors, such as match location, toss outcomes, and recent team form, were considered to enhance the predictive power of the model. Best practices from leading AI and data science platforms guided the technical implementation, ensuring that the model adhered to established machine learning principles. Various machine learning models, including logistic regression, decision trees, random forests, and gradient boosting, were explored to optimize predictive accuracy. Each algorithm was tested and compared based on its ability to analyze past data and predict future match outcomes. Hyperparameter tuning and cross-validation techniques were applied to refine model performance and reduce overfitting_([Decision Tree, Random Forest, and XGBoost: An Exploration into the Heart of Machine Learning, 2024](#)). Performance evaluation metrics including accuracy, precision, recall, and F1-score were used to assess model reliability and effectiveness in real-world applications. The IoA Competency Framework principles were integrated throughout these processes to ensure adherence to industry standards and maintain consistency in analytical rigour. Ethical and contextual considerations played a vital role in model development. Bias mitigation strategies were implemented to prevent skewed predictions, ensuring fairness and transparency in results. The study acknowledged the potential limitations of machine learning models in sports analytics, particularly the unpredictability of human decision-making, external conditions, and in-game dynamics. To address these challenges, external factors such as weather conditions, player injuries, and match-day strategies were systematically analyzed and integrated into the model. These variables added another layer of accuracy, making the predictions more robust and reflective of real-world scenarios.

A structured project lifecycle was followed, including data preprocessing, exploratory analysis, model selection, training, evaluation, and iterative refinement. Data preprocessing involved cleaning and normalizing datasets to eliminate inconsistencies and missing values. Exploratory analysis helped uncover correlations between different features, guiding model selection and feature importance ranking. The training phase involved feeding historical match data into the selected models and optimizing them for the best possible accuracy. The evaluation was conducted using both training and test datasets, ensuring that the model performed reliably on unseen data. Continuous monitoring and validation ensured that the model dynamically adapted to new data and evolving match conditions_([The Ultimate Guide to Feature Monitoring for Real-Time Machine Learning, 2023](#)). As the IPL format evolves and player rosters change each season, the model was designed to be flexible and scalable, allowing for real-time updates. Post-model deployment, regular updates and retraining would be conducted to improve accuracy and account for recent performances and trends ([Komolafe, 2025](#)). This research establishes a reliable framework for IPL match prediction by applying a rigorous and systematic approach, contributing to advancements in sports analytics and decision-making. The findings of this study have practical applications for teams, analysts, and fans, offering data-driven insights that can enhance strategic planning

and match preparations. The integration of machine learning in sports analytics continues to grow, and this research adds value by demonstrating the effectiveness of predictive models in professional cricket.

Case Studies:

IBM - Predictive Analytics for Sports Performance

Background

International Business Machines Corporation (IBM) is a globally renowned technology company known for its artificial intelligence (AI), cloud computing, and data analytics innovations. Since its establishment in 1911, IBM has continuously evolved to meet the changing needs of healthcare, finance, education, and telecommunications industries. One of IBM's most significant contributions in recent years has been in AI through IBM Watson. This sophisticated AI system utilizes machine learning and natural language processing to solve complex problems. Among the many areas where IBM's AI has impacted, professional sports stand out as a particularly compelling application ([VanderLinden, n.d.](#)). In collaboration with the Australian Cricket Board, IBM launched a pioneering initiative to use predictive analytics and machine learning to enhance team performance and game strategy. The system used historical match data, player statistics, and psychological traits to analyze player potential and match outcomes comprehensively. By analyzing key metrics such as batting averages, bowling rates, injury history, mental toughness, and environmental factors like weather and pitch conditions, IBM's model allowed coaches to make data-driven decisions on team selection, field placements, and in-game strategies. This approach moved away from intuition-based decisions and brought a new era of scientific precision to the game. The success of this initiative highlighted the potential of predictive analytics in sports and laid the groundwork for its application in high-stakes tournaments like the Indian Premier League (IPL), where accurate performance forecasting can provide a critical competitive edge. IBM's predictive analytics system in cricket has been successful for several compelling reasons. First and foremost, it allowed for a data-driven approach to strategy, replacing traditional reliance on experience or intuition. Coaches could make informed decisions using various data points, from statistical performance metrics to psychological evaluations.

Ethical Issues

Despite its advantages, using predictive analytics in sports by companies like IBM raises important ethical concerns. One major issue is player privacy. Collecting sensitive data, including physical health, psychological profiles, and injury records, requires informed consent and careful data management ([Awale, 2023](#)). Players may feel surveilled or exposed without transparency, leading to discomfort or distrust. Additionally, there is a risk of bias in machine learning models. If the data used to train the models is skewed or incomplete, the predictions might favor certain players or unfairly disadvantage others, thereby perpetuating inequality in team selections. Another concern is the lack of transparency and accountability in AI systems. Coaches and management might rely heavily on algorithmic

recommendations without fully understanding how these decisions are made, leading to a detachment from human judgment. The constant evaluation and surveillance could also psychologically impact players, increasing stress and reducing performance. Furthermore, if some teams have access to more advanced analytics tools than others, it may create an imbalance in competition, undermining the fairness and integrity of the sport. These ethical challenges must be carefully managed to ensure that the benefits of predictive analytics do not come at the cost of player rights and competitive fairness.



Figure 9 IBM - Predictive Analytics for Sports Performance

Reasons for Success

IBM's predictive analytics system in cricket has been successful for several compelling reasons. First and foremost, it allowed for a data-driven approach to strategy, replacing traditional reliance on experience or intuition. Coaches could make informed decisions using various data points, from statistical performance metrics to psychological evaluations. Integrating physical and mental aspects of player performance gave the system a holistic edge, enabling it to evaluate players more accurately (IBM, 2022). Real-time data processing capabilities enhance decision-making during matches, allowing coaches to dynamically adapt strategies based on unfolding events, such as player fatigue, weather changes, or scoring momentum. Another strength of the system was its adaptability to different

conditions. Considering variables like pitch type, opposition strength, and weather forecasts helped teams customize their strategies to specific scenarios. This level of precision contributed to more efficient team selection, improved coordination, and smarter tactical decisions. Moreover, the system's scalability means that it can be applied in international cricket and league-based competitions like the IPL, where it can offer a significant strategic advantage in a tightly contested environment.

Possible Reasons for Failure

Despite its success, IBM's predictive analytics system may face challenges that could hinder its effectiveness. One significant risk lies in the quality and availability of data. If data is incomplete, inaccurate, or inconsistent, the predictions generated by the system may be unreliable or misleading. Overreliance on technology is another concern. When coaches and managers depend too heavily on AI recommendations, they may overlook contextual factors or human intuition that are still essential in sports. Additionally, the complexity of AI models can make them difficult for non-technical users to interpret, which may result in misunderstandings or misapplication of the insights [\(IBM, 2024\)](#). The unpredictable nature of sports is also a limiting factor. Unexpected events such as injuries, sudden changes in form, or unique player behaviors can defy even the most sophisticated predictive models. Resistance to change among traditional coaches or sports organizations may also pose a barrier, as they may be hesitant to trust or adopt new technologies. Finally, legal, and ethical limitations, such as data protection regulations and player consent requirements, could restrict the full implementation of these systems. These factors highlight the importance of balancing technological innovation with practical and ethical considerations to ensure long-term success.

Opta Sports - Performance Analysis and Prediction Models

Background

Opta Sports is a global leader in sports data analytics, known for its advanced breakdown of team and individual player performance across various sports, particularly cricket. Since its establishment in 1996, Opta has transformed how sports professionals, analysts, and fans understand the nuances of competitive matches. Their main strength is collecting and analyzing large volumes of real-time data, including player movements, run rates, fielding positions, batting and bowling performance, and contextual elements like pitch condition and weather [\(Arastey, Sport Performance Analysis, June \)](#). Opta's systems integrate this data with powerful machine learning models, such as random forest and regression algorithms, to create predictive tools capable of forecasting match outcomes with a high degree of accuracy. These tools are especially relevant in fast-paced, unpredictable tournaments like the Indian Premier League (IPL), where coaches and analysts depend on real-time insights to make split-second decisions.

Ethical Issues

Despite its technological advancements, Opta's data-driven approach brings specific ethical concerns. The collection and use of detailed player data including physical performance, psychological patterns, and emotional responses raises questions about player privacy and informed consent. If this sensitive data is used without explicit permission or is shared irresponsibly with third parties, it may lead to misuse or exploitation (AmityCrestwood, 2023). Furthermore, using predictive analytics in live matches could lead to manipulation or unfair advantages if team access to insights is not equally distributed. There is also the risk of players being reduced to data points, with decisions about selection or tactics based solely on algorithmic outputs rather than human intuition or contextual judgment. Such practices can create ethical dilemmas around fairness, transparency, and accountability in sports.



Figure 10 Opta Sports - Performance Analysis and Prediction Models

Reasons for Success

Opta's success can be attributed to its ability to combine vast datasets with sophisticated machine learning algorithms to generate accurate, context-sensitive predictions. Their models are tailored to the specific requirements of different competitions, such as the IPL, where they account for unique variables like pitch behavior, player form, opposition dynamics, and crowd influence [\(Hugh, 2025 \)](#). The inclusion of real-time data updates allows analysts and coaches to adapt strategies mid-game, making decisions like rotating bowlers or changing field placements based on live projections. Opta also adds value beyond teams, media broadcasters, and commentators, who use their data to enhance the fan experience with engaging and insightful match analysis. Opta has positioned itself as an indispensable tool in modern cricket strategy and sports broadcasting by delivering actionable insights at both pre-match and in-game stages.

Possible Reasons for Failure

While Opta's systems are highly advanced, several factors could limit their effectiveness. One primary concern is over-dependence on data models, which may not always account for spontaneous, human elements of the game, such as unpredictable player behavior, sudden injuries, or emotional resilience under pressure. Inaccurate or biased data can also lead to faulty predictions, especially if models are not regularly updated to reflect current performance trends [\(Arastey, Artificial Intelligence \(AI\) in Sports, 2021\)](#). Technical failures in real-time data collection, such as sensor errors or connectivity issues, can result in misleading analytics during crucial match phases. Additionally, misinterpreting data by coaches or analysts without proper training could lead to poor decision-making. Lastly, if ethical issues like privacy concerns or unequal data access are not addressed, trust in the system could erode, potentially leading teams or leagues to limit or abandon their use.

Sportradar - AI and Machine Learning for Predictive Analytics and Sports Betting Systems

Background

Sportradar, founded in 2001, is a global leader in the sports analytics and data technology industry, delivering cutting-edge solutions to clients ranging from betting companies and broadcasters to teams and sports organizations. With a strong focus on artificial intelligence (AI) and machine learning (ML), Sportradar provides real-time data feeds, performance analytics, and predictive modeling across various sports, including cricket, football, basketball, and tennis. In the context of the Indian Premier League (IPL), Sportradar applies AI-driven analysis on historical and real-time data to predict match outcomes, assess player performance, and offer deep strategic insights [\(THE SPORTS TECHNOLOGY COMPANY, n.d.\)](#). By analyzing past match data, current player form, team strategies, and in-game dynamics, the company helps stakeholders make informed decisions before and during matches. Their

services are particularly valuable in high-velocity formats like the IPL, where real-time analytics can guide on-the-spot decisions for teams and betting platforms.

Ethical Issues

While Sport radar's technology offers immense value, it raises several ethical concerns. Detailed player data often in real time raises questions regarding data ownership, privacy, and consent. Players might not always know how deeply their performance metrics are tracked or used, particularly by third parties like betting companies. Additionally, the close relationship between Sport radar's predictive systems and sports betting introduces ethical challenges, as it risks reinforcing gambling behavior or being misused for match-fixing and insider betting. Data bias or misinterpretation is also risky, which can unfairly impact player evaluations or betting outcomes [\(Shah, 2025\)](#). Without proper oversight, the widespread use of predictive analytics can commodify athletes, reducing them to algorithmic profiles rather than human competitors.



Figure 11 Sport radar -Predictive Analytics and Sports Betting Systems

Reasons for Success

Sportradar's success lies in its ability to process massive volumes of historical and live data using advanced machine learning models to produce highly accurate predictions. Their dynamic algorithms are constantly refined by real-time inputs, improving the precision of forecasts over time. This adaptability makes their systems particularly effective in fast-paced leagues like the IPL. They provide stakeholders, including coaches, analysts, broadcasters, and bookmakers, actionable insights on player performance, team strategies, and even in-match tactics [\(Tshimula, 2024\)](#). Their predictive models enhance the live sports experience by enabling real-time commentary, smarter betting odds, and data-driven coaching. Sportradar's reach across multiple sports and markets has also allowed it to build a robust and scalable infrastructure, making it a dominant force in sports analytics globally.

Possible Reasons for Failure

Despite its technological strengths, Sportradar could face challenges impacting its reliability and reputation. One potential failure point is the system's over-reliance on data, which might overlook unpredictable human factors such as pressure handling, sudden injuries, or emotional states. Technical errors during data collection or incorrect modeling assumptions could lead to inaccurate predictions, especially if models are not regularly updated or tested. There is also a risk of stakeholder overconfidence in AI predictions, potentially sidelining expert intuition or strategic flexibility. Ethical backlash from excessive involvement in gambling or privacy concerns could also damage the company's credibility [_\(Butt, 2024\)](#). Moreover, if competitors develop more transparent or player-friendly analytics systems, Sportradar might struggle to maintain its leadership position in a rapidly evolving market.

Integration

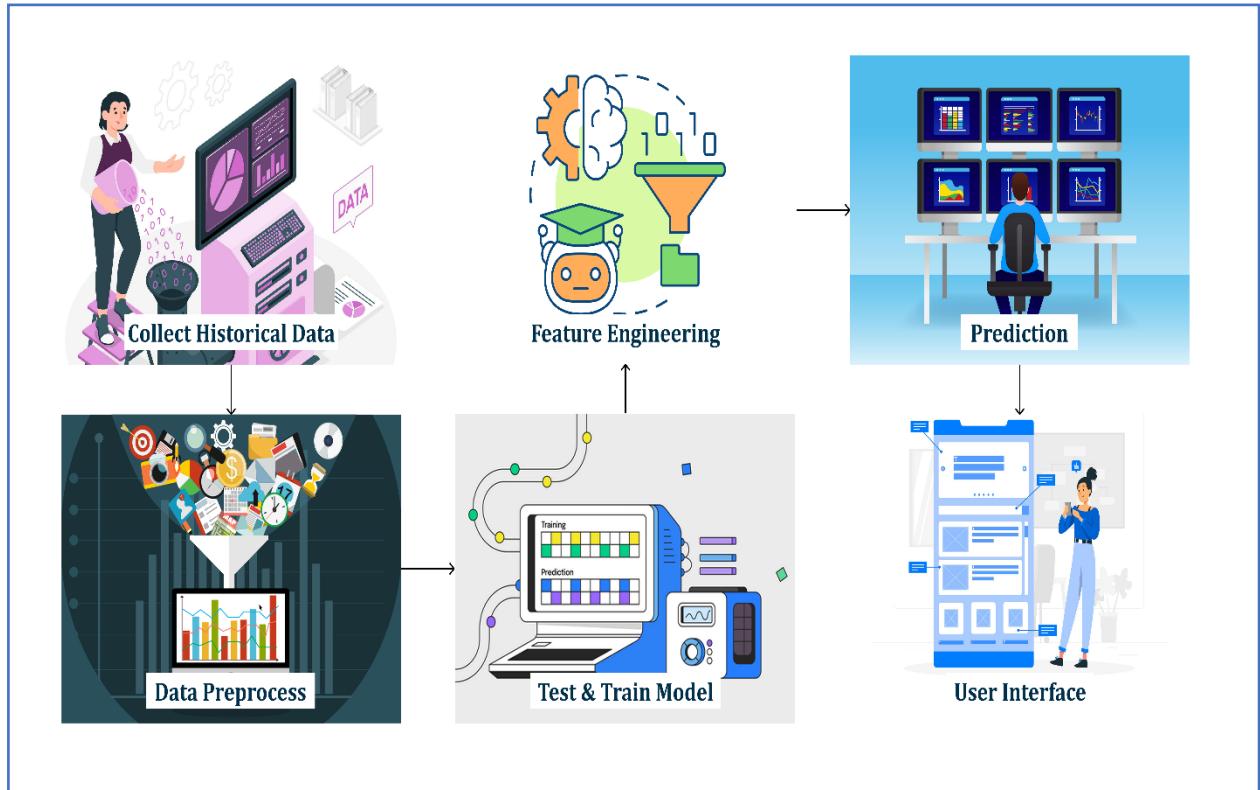


Figure 12 Integration of the model

System integration is an end-to-end machine learning pipeline from raw match data to actionable and accurate predictive information. Constructing such a system begins with harvesting good-quality structured data through Kaggle, a popular open-source platform widely followed in the data science process. The data on which models are run are the nitty-gritty historical facts of IPL matches, such as match results, player and team statistics, stadium details, toss outcomes, etc., and context-specific characteristics. Following data collection, the data pre-processing is to the extent that the dataset is clean and ready for processing. This encompasses handling missing values, deduplication of records, normalization to a standard format, and converting categorical variables to numeric format through label encoding or one-hot encoding [\[Sethi, 2025\]](#). This is a critical step so that there is integrity and consistency throughout the dataset, and this has a direct impact on the performance of the model. The second most crucial step is feature engineering, in which domain and relevant features are built to enhance the model's predictive ability. In cricket, specifically T20 games such as the IPL, there exist some dynamic features that have a

significant contribution towards deciding the game. These are the run rate till now (runs till now scored by the team per over taken), required run rate (runs remaining to be scored by the team per over left), wickets fallen, balls left in the innings, and wickets left. These work especially well for real-time prediction of live matches or simulations since they can understand the dynamic situation of the match conditions and allow the model to make more situation-specific choices. Once the feature set is determined, the data is split into test sets and training sets, usually in the ratio of 80:20 or 70:30. Training data is utilized for training various machine learning models such as Decision Trees, Random Forest, Logistic Regression, and XGBoost [\(Galarnyk, 2025\)](#). The model is also tested and cross-validated with some of the performance metrics such as accuracy, precision, recall, and F1-score to identify what worked well using which approach. Hyperparameter tuning and cross-validation techniques are used to enhance the capability of the model to generalize as well as prevent overfitting excessively [\(Omarzai, 2024\)](#). After the detection of the highest-performing model, user access and deployment represent the last step of integration. A web-based interface is crafted using Streamlit, an open-source Python library for building interactive web applications. The interface is used to let users input match-specific parameters e.g., current score, overs left, wickets down, and runs remaining to win to receive an estimated match result. The interface is responsive, simple, and clean, and therefore, both technical and non-technical users can use it effortlessly. Due to this end-to-end integration, the system offers an operational data-driven solution to machine learning-based prediction of IPL match outcomes.

Findings

1.What are the most effective machine learning algorithms used for predicting IPL match outcomes, and how do they compare in terms of accuracy, interpretability, and computational efficiency?

Predicting IPL match outcomes using machine learning involves leveraging various algorithms, each with unique strengths and weaknesses in accuracy, interpretability, and computational efficiency. Random Forest is one of the most accurate models used in this context. It is an ensemble learning technique that builds multiple decision trees and aggregates their results to produce more reliable predictions [\(IBM, 2025\)](#). Its ability to handle categorical and numerical data makes it suitable for analyzing player statistics, team performance, and match conditions. However, despite its high accuracy, Random Forest is often criticized for its lack of interpretability being a "black-box" model makes it difficult to understand the reasoning behind its predictions. Additionally, training Random Forest models on large datasets can be computationally expensive, making them less suitable for real-time applications. Another widely used model is XGBoost (Extreme Gradient Boosting), which offers improved performance through regularization techniques and parallel processing. It is known for its high accuracy and efficiency, especially when working with large datasets that contain numerous features. XGBoost typically outperforms Random

Forest regarding speed and predictive power, making it a strong choice for IPL outcome prediction ([Ashtari, 2024](#)). However, like Random Forest, it lacks interpretability, which can be a drawback when transparency in decision-making is required by stakeholders such as coaches or analysts.

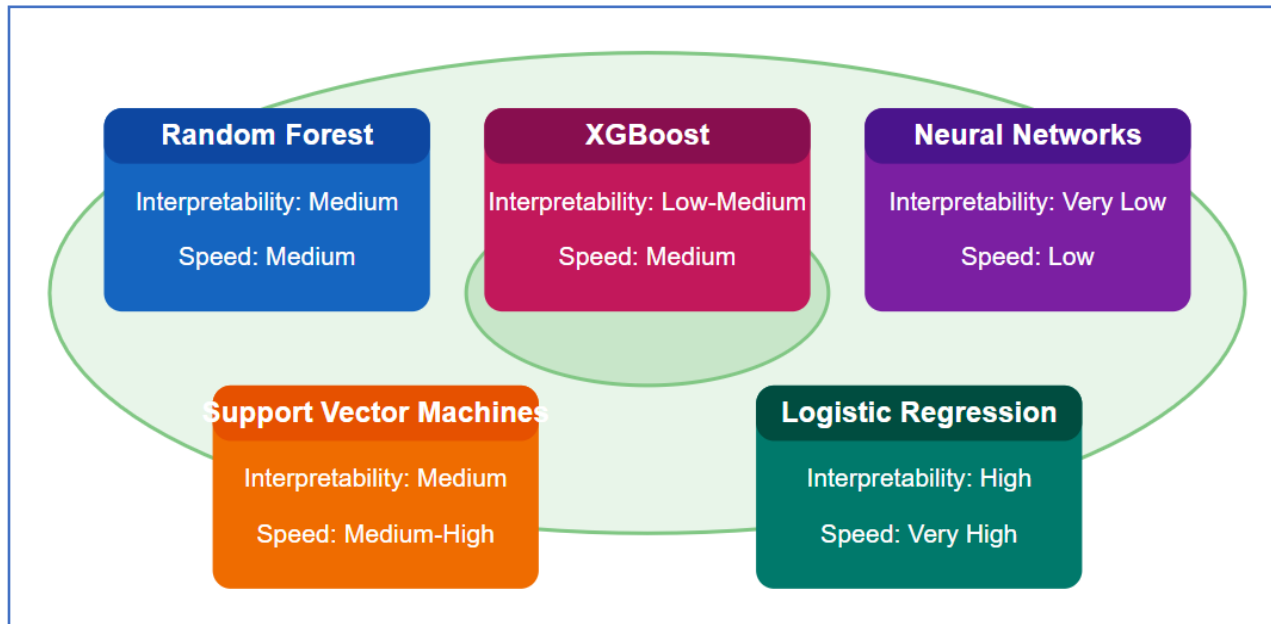


Figure 13 Most effective machine learning algorithms used for predicting IPL

When interpretability is a priority, Logistic Regression is often the model of choice. It is particularly effective for binary classification problems, such as predicting whether a team will win or lose. Unlike ensemble models, Logistic Regression explains how each input feature, such as batting averages, pitch type, or weather, affects the final prediction. While it may not match the predictive accuracy of more complex models, its simplicity allows faster computation and easier deployment in real-time systems ([Amit, 2025](#)). However, it struggles with complex, non-linear relationships in data. Other models, such as Support Vector Machines (SVM) and Neural Networks, are also used in IPL prediction. SVMs perform well with high-dimensional data and smaller datasets, but they require intensive tuning and are not well-suited to noisy or massive datasets. Neural Networks, on the other hand, excel at capturing intricate, non-linear patterns in large and diverse datasets ([Syed Asghar Ali Shah, 2025](#)). They are highly accurate, require significant computational resources, and are almost entirely opaque in their decision-making process. The best approach depends on the specific priorities of the use case. If accuracy is the primary concern and resources are available, XGBoost offers the best performance. If interpretability is crucial, Logistic Regression is the most transparent and efficient. For a balanced approach, Random Forest provides a firm middle ground, offering high accuracy with moderate computational demands, though at the cost of reduced interpretability.

2. What are the ethical and data privacy challenges associated with using machine learning for IPL winner prediction, and how can transparency and fairness be ensured in predictive sports analytics?

The increasing use of machine learning (ML) and artificial intelligence (AI) in sports analytics, particularly in predicting outcomes like the winners of Indian Premier League (IPL) matches, introduces not only technical advancements but also a host of ethical and data privacy concerns [\(Carrio, 2023\)](#). As these predictive systems become more accurate and influential in shaping betting odds, coaching decisions, and even public expectations, it becomes crucial to address how such systems are developed and used ethically and fairly. Moreover, with large volumes of data being collected, stored, and processed, ensuring user and player privacy is essential. One of the primary ethical challenges in using ML for IPL winner prediction lies in the fairness and bias of the models. Machine learning algorithms are only as good as the data they are trained on. The predictions will reflect these biases if the historical data used to train these models is biased or unbalanced—for example, if it underrepresents specific teams, stadiums, or conditions. This could unfairly disadvantage specific teams or create inaccurate public narratives. Moreover, the black-box nature of many ML models, such as deep neural networks or ensemble methods like Random Forest and XGBoost, makes it difficult to interpret why specific predictions were made. Without interpretability, detecting or correcting such biases becomes nearly impossible, thereby undermining the fairness of the predictive process. Another ethical issue involves the potential misuse of predictions. Advanced models can give betting platforms and stakeholders access to highly accurate forecasts, raising concerns about match-fixing or manipulating betting markets. Suppose powerful predictive tools are only available to a select few, such as betting syndicates or privileged insiders. In that case, it can create an uneven playing field that erodes public trust in the integrity of the sport. This is particularly sensitive in the context of the IPL, where betting is highly popular and has a history of controversy. Regarding data privacy, collecting player statistics, health records, behavioral data, and even real-time GPS or biometric information raises serious concerns. At the same time, match statistics are usually public, personal, and sensitive data, such as a player's injury history, mental health conditions, or internal team strategies, but they are not. Using such data without consent breaches privacy rights and can expose players to scrutiny or stress that affects their performance and well-being. Sometimes, even anonymized data can be reverse engineered to re-identify individuals, further complicating the issue [\(SAP, 2024\)](#). Several strategies must be employed to ensure transparency and fairness. Firstly, predictive models should prioritize explainability. Explainable AI (XAI) techniques should be integrated into model design to allow stakeholders to understand how predictions are being made. Tools like SHAP (Shapley Additive Explanations) or LIME (Local Interpretable Model-agnostic Explanations) can help identify which features most influence a prediction. This makes it easier to detect bias, build trust among users, and provide feedback for continuous improvement. Data governance policies should be established and enforced. These policies must define what data can be collected, how it should be anonymized, who can access it, and under what conditions it can be used. Players should give consent before using any private or sensitive data. Data minimization principles collecting only the data necessary for the intended purpose should be followed to reduce the risk of privacy breaches.

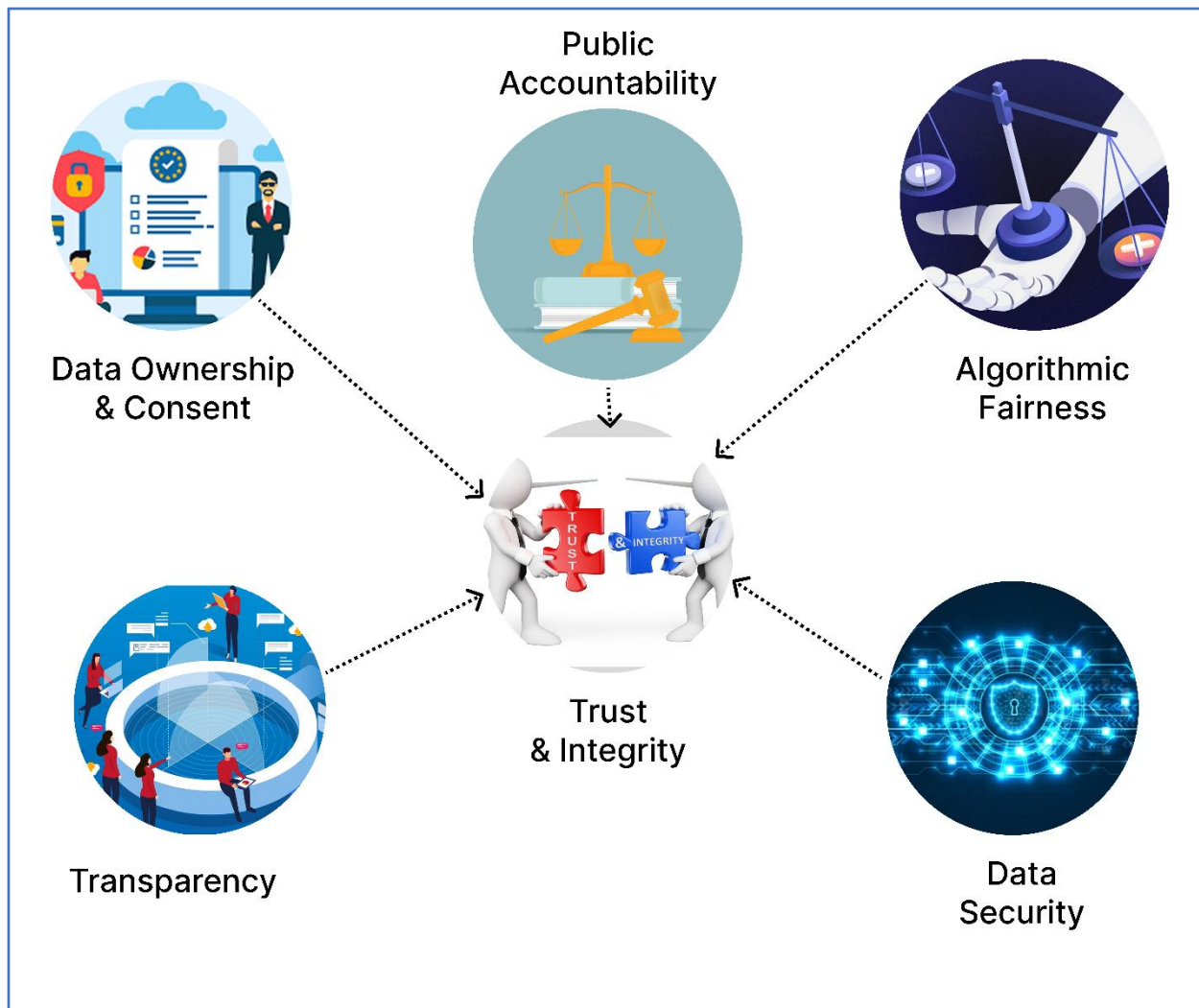


Figure 14 Ethical and privacy considerations analyzing match data

Regulatory oversight also plays an important role. In collaboration with technology providers, sports regulatory bodies should define ethical guidelines and ensure that all stakeholders follow them. This includes transparency in disclosing which algorithms are used, how they are trained, and how the results are utilized [\(Pantelis Linardatos, 2021\)](#). Ethical audits and third-party evaluations should be conducted periodically to assess whether these systems operate fairly and within legal boundaries. Education and awareness are additional keys to building transparency and fairness. Coaches, players, analysts, and even fans should understand how ML predictions are generated. If stakeholders understand the strengths and limitations of predictive systems, they can make more informed decisions and resist over-reliance on algorithmic forecasts. The efforts must be made to promote inclusivity and accessibility in predictive analytics. Tools and insights derived from ML should not be limited to extensive franchises or powerful betting platforms. Equal access ensures that all teams and stakeholders benefit from technological advancements, promoting fairness across the league. While machine learning offers immense potential for improving the accuracy and excitement of IPL winner predictions, it must be implemented responsibly. By addressing ethical concerns such as fairness, bias, and misuse of predictions,

and by safeguarding player data through strong privacy protections, the industry can build predictive systems that are not only powerful but also transparent, fair, and trustworthy. Responsible AI in sports analytics will maintain the integrity of competitions like the IPL and ensure that technological innovation serves everyone involved in the game.

Future Works

In future research, the IPL winner prediction system can be further enhanced by incorporating a broader spectrum of contextual and real-time variables, such as live weather data, pitch conditions, last-minute player substitutions, injury reports, and captaincy decisions. These dynamic elements, although available through modern broadcasting and data feeds, are often underutilized in predictive models. Including them could significantly improve the model's real-time adaptability, relevance, and accuracy. Furthermore, expanding the dataset to encompass a longer history of IPL seasons, while ensuring data quality and consistency, would help capture evolving team strategies, player performance trends, and rule changes over time. Integrating diverse data sources such as social media sentiment analysis, fan engagement metrics, betting odds, and expert panel insights can help capture subtle psychological and environmental factors that influence match outcomes. To make the system faster and more reliable, future implementations should leverage optimized data pipelines and cloud-based big data platforms such as AWS, Google Cloud, or Azure. Distributed processing tools like Apache Spark or Kafka can facilitate the real-time ingestion and processing of massive datasets. On the modeling side, the deployment of cutting-edge deep learning architectures, including Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models, can help capture temporal dependencies and sequential patterns inherent in match progressions.



Figure 15 Future Works of the Project

Additionally, ensemble models combining the strengths of traditional machine learning (e.g., Random Forest, XGBoost) and deep learning can further boost robustness.

Model interpretability will remain a central challenge, especially in complex architectures. Hence, incorporating Explainable AI (XAI) techniques such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), and integrated gradient methods will be vital to ensure transparency, user trust, and model auditing. Ethical considerations must also be prioritized in future enhancements—this includes implementing stricter data privacy protocols, ensuring proper user consent for data use, preventing misuse for gambling or match-fixing, and eliminating algorithmic biases through fairness-aware training methods. The system's usability can also be improved through the development of intuitive user interfaces and dashboards using platforms like Streamlit, Dash, or Tableau. These should support scenario-based simulations, allowing users to adjust match conditions and immediately see predicted outcomes. Incorporating voice-based or chatbot interfaces could also make the system more interactive. Finally, integrating the model into mobile applications or sports analytics platforms via APIs will ensure broader accessibility, enabling stakeholders, including teams, analysts, broadcasters, and fans, to access predictions in real-time, thereby maximizing the system's practical impact in the evolving sports analytics landscape.

Conclusion

This study has systematically investigated the application of machine learning techniques for predicting the outcomes of Indian Premier League (IPL) matches. By analyzing historical match data, player performance metrics, and team statistics, the research has demonstrated that data-driven models can provide meaningful insights into match outcomes. Algorithms such as decision trees, random forests, and logistic regression were implemented and evaluated, revealing that while traditional models achieve a reasonable level of accuracy, there remains ample room for enhancement through more sophisticated approaches. The study also identified key factors that significantly influence match predictions, underlining the importance of comprehensive data preprocessing and thoughtful feature engineering. Furthermore, the research underscores the necessity of integrating contextual and real-time variables, such as live weather updates, player injuries, and team dynamics, to improve prediction reliability. Ethical considerations, including data privacy, algorithmic fairness, and transparency, were also highlighted as essential components of responsible AI development. The use of tools like Streamlit for visualization and user interaction illustrates how predictive systems can be made accessible, interactive, and user-friendly for a broad audience, including analysts, researchers, and fans. This thesis not only showcases the feasibility and effectiveness of using machine learning in sports analytics but also lays the groundwork for future innovations in the field. With the continuous evolution of artificial intelligence, richer data sources, and more powerful computational tools, IPL prediction systems can be further refined to deliver faster, more accurate, and more interpretable outcomes, ultimately transforming how fans, coaches, and analysts engage with the game of cricket.

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Appendix

Links

GitHub Link: https://github.com/Nitesh511/IPL_Winner_Prediction

Video Link: <https://youtu.be/g4tzMYklu0k>

Project Plan

Task	Start Date	Duration (days)	Status	End Date
Topic Finalization and Proposal Approval, Literature Review and Problem Statement	12/02/2024	20	Done	12/22/2024
Data Collection (Kaggle, APIs, Sources),	12/22/2024	16	Done	01/07/2025
Develop a machine learning model which can predict IPL winner prediction using Historical Data	01/07/2025	23	Done	01/30/2025
Feature Engineering and Selection, Model Evaluation and Optimization Visualization & Dashboard (Streamlit)	01/30/2025	21	Done	02/20/2025
Document the research process, model development and results	02/20/2025	75	Done	05/05/2025
Review & Feedback	05/05/2025	3	Done	05/08/2025
Thesis Submission and Conclusion	05/08/2025	4	Done	05/12/2025

Figure 16 Project Plan

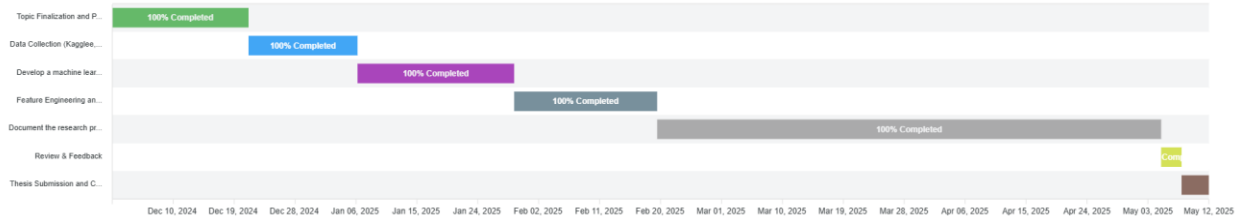


Figure 17 Gantt Chart

Issue Logs

Issue Category	Issue	Status	Likelihood	Impact	Mitigation Procedure
Ethical	Potential bias in player or team predictions	Open	High	High	Implement XAI techniques and diverse datasets to reduce bias.
Data Quality	Incomplete or noisy historical match data	Closed	High	High	Use data preprocessing, cleaning techniques, and reliable data sources.
Experience & Skill	Limited knowledge in deep learning models	Closed	Medium	High	Participate in advanced ML courses and consult with domain experts.
Integration	Real-time data feed integration issues	Open	Medium	Medium	Use well-documented APIs and conduct integration testing.
Security	API keys or model endpoints being exposed	Closed	Low	High	Secured API keys and restricted endpoint access.
Scalability	Model performance drops with larger data	Open	Medium	Medium	Apply model optimization and cloud-based scaling strategies.
Time Management	Risk of delay in thesis writing and testing	Open	Medium	High	Create a Gantt chart and stick to a strict project timeline.
Performance	Slow model prediction during real-time tests	Open	Medium	Medium	Optimize model and use GPU support if necessary.
Legal Compliance	Use of copyrighted or restricted data	Closed	Low	Medium	Ensured data licensing and proper citation.

Figure 18 Issue Logs

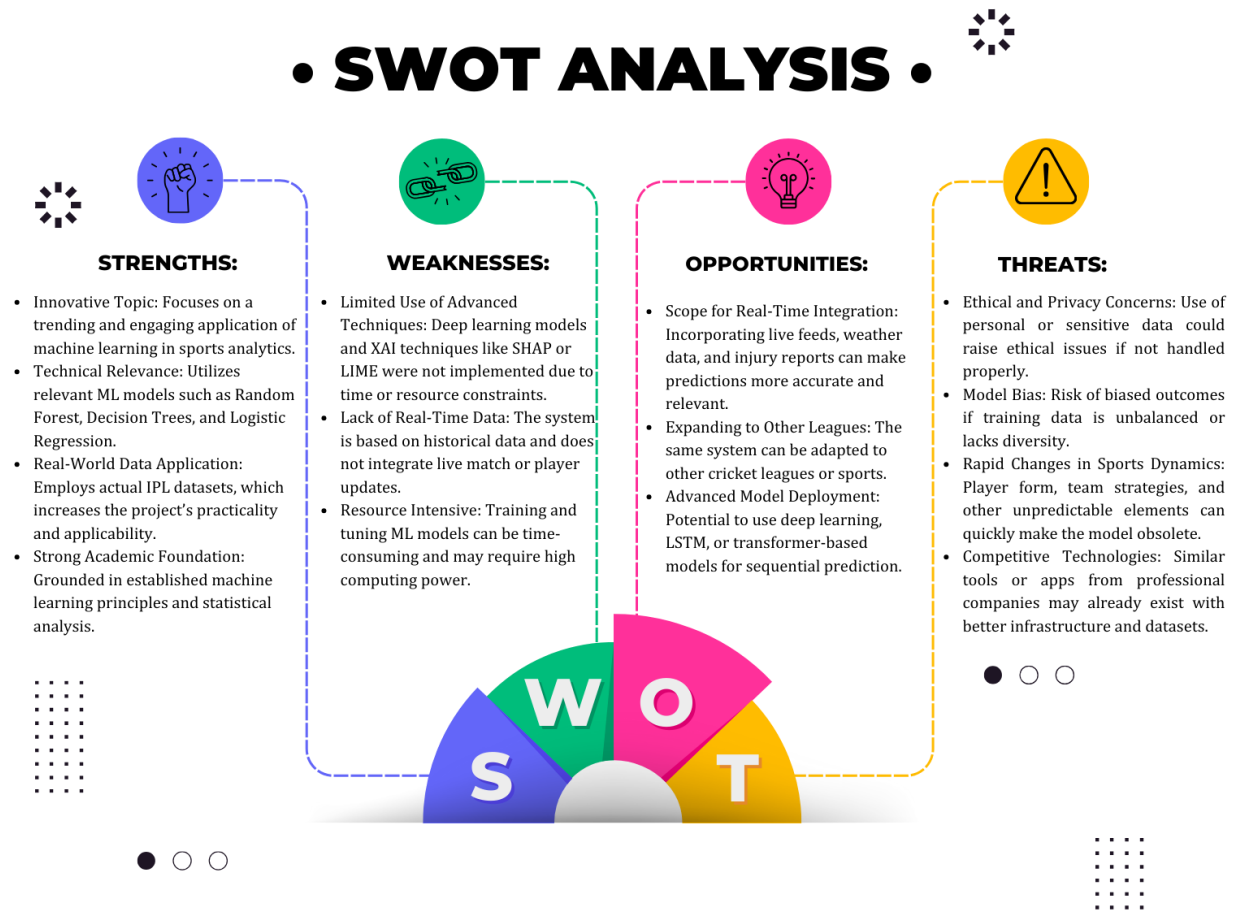


Figure 19 Swot Analysis of the project

Screenshots

IPL Match Winner Predictor

[Match Predictor](#) [About the Model](#)

Batting Team 	Bowling Team 
Chennai Super Kings	Delhi Capitals
Target Score 	Match Venue 
170	Abu Dhabi
Current Score 	Overs Completed 
100	12.00
Wickets Lost 	
3	

Predict Match Outcome 

70

Runs Needed

48.0

Balls Left

7

Wickets in Hand

8.33

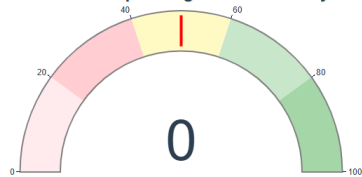
Current RR

8.75

Required RR

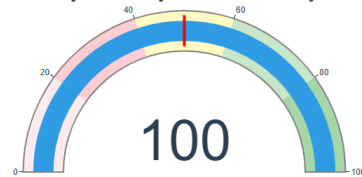
Rajasthan Royals has a 100.0% chance of winning!

Chennai Super Kings Win Probability



R

Rajasthan Royals Win Probability



Match Analysis & Insights

Run Rate Comparison



Match Progression



How Chennai Super Kings's Win Probability Could Change



Key Factors Influencing the Prediction

Factors Favoring Chennai Super Kings:

- With 7 wickets remaining, Chennai Super Kings has good batting depth
- Chennai Super Kings: Masters of chasing targets with composed batting line-up

Factors Favoring Rajasthan Royals:

- Needing 70 runs from 48.0 balls requires more than a run a ball
- Rajasthan Royals: Tactically astute with focus on role clarity

Other Match Factors:

- Required run rate (8.75) is slightly higher than current run rate (8.33) - pressure on batting side

```
In [74]: # Importing the pandas library for data manipulation and analysis
import pandas as pd

# Importing the numpy library for numerical operations
import numpy as np
# Importing the matplotlib library for Graph
import matplotlib.pyplot as plt
```

```
In [75]: # Reading the CSV file named 'matches.csv' from the 'Data' folder into a DataFrame
match= pd.read_csv('Data/matches.csv')
# Displaying the first five rows of the DataFrame to get an overview of the dataset
match.head()
```

```
Out[75]:
```

	id	season	city	date	match_type	player_of_match	venue	team1	team2	toss_winner	toss_decision	winner	result	re
0	335982	2007/08	Bangalore	2008-04-18	League	BB McCullum	Chinnaswamy Stadium	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	field	Kolkata Knight Riders	runs	
1	335983	2007/08	Chandigarh	2008-04-19	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings	bat	Chennai Super Kings	runs	
2	335984	2007/08	Delhi	2008-04-19	League	MF Maharoof	Feroz Shah Kotla	Delhi Daredevils	Rajasthan Royals	Rajasthan Royals	bat	Delhi Daredevils	wickets	
3	335985	2007/08	Mumbai	2008-04-20	League	MV Boucher	Wankhede Stadium	Mumbai Indians	Royal Challengers Bangalore	Mumbai Indians	bat	Royal Challengers Bangalore	wickets	
4	335986	2007/08	Kolkata	2008-04-20	League	DJ Hussey	Eden Gardens	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	bat	Kolkata Knight Riders	wickets	

```
In [76]: # Reading the CSV file named 'deliveries.csv' from the 'Data' folder into a DataFrame
delivery=pd.read_csv('Data/deliveries.csv')
# Displaying the first five rows of the DataFrame to get an overview of the dataset
delivery.head()
```

```
Out[76]:
```

	match_id	inning	batting_team	bowling_team	over	ball	batter	bowler	non_striker	batsman_runs	extra_runs	total_runs	extras_type	is_wicket	playe
0	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	1	SC Ganguly	P Kumar	BB McCullum	0	1	1	legbyes	0	
1	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	2	BB McCullum	P Kumar	SC Ganguly	0	0	0	NaN	0	
2	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	3	BB McCullum	P Kumar	SC Ganguly	0	1	1	wides	0	
3	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	4	BB McCullum	P Kumar	SC Ganguly	0	0	0	NaN	0	
4	335982	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	5	BB McCullum	P Kumar	SC Ganguly	0	0	0	NaN	0	

```
In [77]: # Displaying the column names of the 'delivery' DataFrame
delivery.columns
```

```
Out[77]: Index(['match_id', 'inning', 'batting_team', 'bowling_team', 'over', 'ball',
               'batter', 'bowler', 'non_striker', 'batsman_runs', 'extra_runs',
               'total_runs', 'extras_type', 'is_wicket', 'player_dismissed',
               'dismissal_kind', 'fielder'],
              dtype='object')
```

```
In [78]: # Displaying the shape (number of rows and columns) of the 'match' DataFrame
match.shape
```

```
Out[78]: (1095, 20)
```



```
In [79]: # Displaying the shape (number of rows and columns) of the 'delivery' DataFrame
delivery.shape
```

```
Out[79]: (260920, 17)
```

```
In [80]: # balling team , batting team , city , run left, ball left, wicket left, total runs, current run rate, required run rate , result
# ...
```

```
In [81]: # Grouping the 'delivery' DataFrame by 'match_id' and 'inning',
# then summing up the 'total_runs' for each group.
total_score_df=delivery.groupby(['match_id','inning']).sum()['total_runs'].reset_index()
```

```
In [82]: total_score_df.head()
```

```
Out[82]:
```

	match_id	inning	total_runs
0	335982	1	222
1	335982	2	82
2	335983	1	240
3	335983	2	207
4	335984	1	129

```
In [83]: # Filtering the 'total_score_df' DataFrame to include only rows where 'inning' is 1
total_score_df=total_score_df[total_score_df['inning']==1]
```

```
In [84]: total_score_df
```

```
Out[84]:
```

	match_id	inning	total_runs
0	335982	1	222
2	335983	1	240
4	335984	1	129
6	335985	1	165
8	335986	1	110
...	...	...	...
2207	1426307	1	214
2209	1426309	1	159
2211	1426310	1	172
2213	1426311	1	175
2215	1426312	1	113

1095 rows x 3 columns

```
In [85]: # Merging the 'match' DataFrame with the 'total_score_df' DataFrame
# using 'match_id' from 'total_score_df' and 'id' from 'match' to combine relevant columns
match_df=match.merge(total_score_df[['match_id', 'total_runs']], left_on='id', right_on='match_id')
```

```
In [86]: match_df.head()
```

```
Out[86]:
```

	id	season	city	date	match_type	player_of_match	venue	team1	team2	toss_winner	...	result	result_margin	target_runs
0	335982	2007/08	Bangalore	2008-04-18	League	BB McCullum	M Chinnaswamy Stadium	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	...	runs	140.0	223.0
1	335983	2007/08	Chandigarh	2008-04-19	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings	...	runs	33.0	241.0
2	335984	2007/08	Delhi	2008-04-19	League	MF Maharoo	Feroz Shah Kotla	Delhi Daredevils	Rajasthan Royals	Rajasthan Royals	...	wickets	9.0	130.0
3	335985	2007/08	Mumbai	2008-04-20	League	MV Boucher	Wankhede Stadium	Mumbai Indians	Royal Challengers Bangalore	Mumbai Indians	...	wickets	5.0	166.0
4	335986	2007/08	Kolkata	2008-04-20	League	DJ Hussey	Eden Gardens	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	...	wickets	5.0	111.0

5 rows x 22 columns

```
In [87]: match_df.columns
```

```
Out[87]:
```

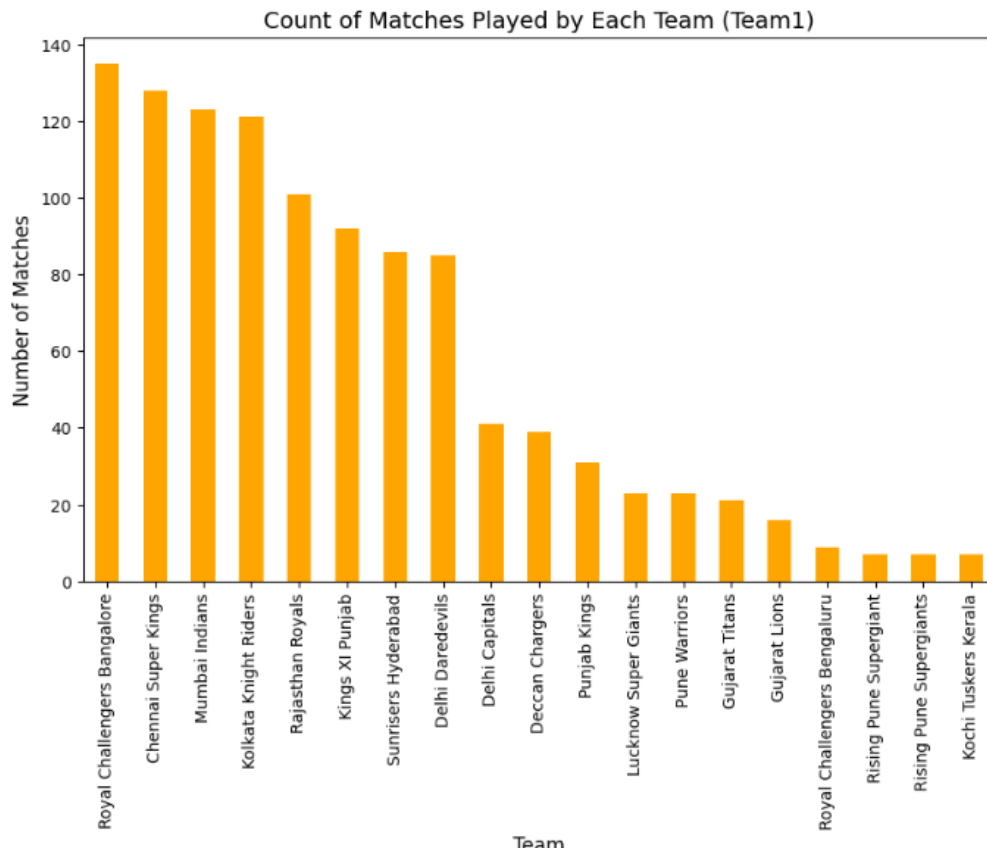
```
Index(['id', 'season', 'city', 'date', 'match_type', 'player_of_match',  
      'venue', 'team1', 'team2', 'toss_winner', 'toss_decision', 'winner',  
      'result', 'result_margin', 'target_runs', 'target_overs', 'super_over',  
      'method', 'umpire1', 'umpire2', 'match_id', 'total_runs'],  
      dtype='object')
```

```
In [88]: # Getting the count of each unique team in the 'team1' column
team_counts = match_df['team1'].value_counts()

# Creating a bar plot for the counts of each unique team
plt.figure(figsize=(10, 6))
team_counts.plot(kind='bar', color='orange')

# Adding titles and labels
plt.title('Count of Matches Played by Each Team (Team1)', fontsize=14)
plt.xlabel('Team', fontsize=12)
plt.ylabel('Number of Matches', fontsize=12)

# Displaying the plot
plt.show()
```



```
In [89]: # Defining a List of teams to work with
teams=['Sunrisers Hyderabad', 'Kolkata Knight Riders', 'Kings XI Punjab', 'Rajasthan Royals', 'Delhi caphtals', 'Chennai Super Ki
'Mumbai Indians', 'Royal Challengers Bangalore']
```

```
In [90]: teams
```

```
Out[90]: ['Sunrisers Hyderabad',
'Kolkata Knight Riders',
'Kings XI Punjab',
'Rajasthan Royals',
'Delhi caphtals',
'Chennai Super Kings',
'Mumbai Indians',
'Royal Challengers Bangalore']
```

```
In [91]: # Replacing 'Delhi Daredevils' with 'Delhi Capitals' in both 'team1' and 'team2' columns
match_df['team1']=match_df['team1'].str.replace('Delhi Daredevils', 'Delhi Capitals')
match_df['team2']=match_df['team2'].str.replace('Delhi Daredevils', 'Delhi Capitals')

# Replacing 'Deccan Chargers' with 'Sunrisers Hyderabad' in both 'team1' and 'team2' columns
match_df['team1']=match_df['team1'].str.replace('Deccan Chargers', 'Sunrisers Hyderabad')
match_df['team2']=match_df['team2'].str.replace('Deccan Chargers', 'Sunrisers Hyderabad')
```

```
In [92]: match_df.head()
```

```
Out[92]:
```

	id	season	city	date	match_type	player_of_match	venue	team1	team2	toss_winner	...	result	result_margin	target_runs
0	335982	2007/08	Bangalore	2008-04-18	League	BB McCullum	M Chinnaswamy Stadium	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	...	runs	140.0	223.0
1	335983	2007/08	Chandigarh	2008-04-19	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings	...	runs	33.0	241.0
2	335984	2007/08	Delhi	2008-04-19	League	MF Maharoof	Feroz Shah Kotla	Delhi Capitals	Rajasthan Royals	Rajasthan Royals	...	wickets	9.0	130.0
3	335985	2007/08	Mumbai	2008-04-20	League	MV Boucher	Wankhede Stadium	Mumbai Indians	Royal Challengers Bangalore	Mumbai Indians	...	wickets	5.0	186.0
4	335986	2007/08	Kolkata	2008-04-20	League	DJ Hussey	Eden Gardens	Kolkata Knight Riders	Sunrisers Hyderabad	Deccan Chargers	...	wickets	5.0	111.0

5 rows x 15 columns

```
In [93]: # Filtering the 'match_df' DataFrame to only include rows where 'team1' is in the list of specified teams
match_df = match_df[match_df['team1'].isin(teams)]

# Filtering the 'match_df' DataFrame to only include rows where 'team2' is in the list of specified teams
match_df = match_df[match_df['team2'].isin(teams)]
```

```
In [94]: match_df.shape
```

```
Out[94]: (621, 22)
```

```
In [95]: match_df.columns
```

```
Out[95]: Index(['id', 'season', 'city', 'date', 'match_type', 'player_of_match',
               'venue', 'team1', 'team2', 'toss_winner', 'toss_decision', 'winner',
               'result', 'result_margin', 'target_runs', 'target_overs', 'super_over',
               'method', 'umpire1', 'umpire2', 'match_id', 'total_runs'],
              dtype='object')
```

```
In [96]: # Selecting specific columns from the 'match_df' DataFrame: 'match_id', 'city', 'winner', and 'total_runs'
match_df = match_df[['match_id', 'city', 'winner', 'total_runs']]
```

```
In [97]: match_df
```

```
Out[97]:
```

	match_id	city	winner	total_runs
0	335982	Bangalore	Kolkata Knight Riders	222
1	335983	Chandigarh	Chennai Super Kings	240
3	335985	Mumbai	Royal Challengers Bangalore	165
4	335986	Kolkata	Kolkata Knight Riders	110
5	335987	Jaipur	Rajasthan Royals	166
...	...	...	...	...
1083	1426298	Kolkata	Kolkata Knight Riders	157
1084	1426299	Chennai	Chennai Super Kings	141
1091	1428309	Ahmedabad	Kolkata Knight Riders	159
1093	1428311	Chennai	Sunrisers Hyderabad	175
1094	1428312	Chennai	Kolkata Knight Riders	113

621 rows x 4 columns

```
In [98]: # Merging the 'match_df' DataFrame with the 'delivery' DataFrame on the 'match_id' column
delivery_df = match_df.merge(delivery, on='match_id')
```

```
In [99]: delivery_df.head()
```

Out[99]:

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	bowler	non_striker	batsman_runs	extra_runs	total_ru
0	335982	Bangalore	Kolkata Knight Riders	222	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	1	SC Ganguly	P Kumar	BB McCullum	0	1	
1	335982	Bangalore	Kolkata Knight Riders	222	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	2	BB McCullum	P Kumar	SC Ganguly	0	0	
2	335982	Bangalore	Kolkata Knight Riders	222	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	3	BB McCullum	P Kumar	SC Ganguly	0	1	
3	335982	Bangalore	Kolkata Knight Riders	222	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	4	BB McCullum	P Kumar	SC Ganguly	0	0	
4	335982	Bangalore	Kolkata Knight Riders	222	1	Kolkata Knight Riders	Royal Challengers Bangalore	0	5	BB McCullum	P Kumar	SC Ganguly	0	0	

```
In [100]: # Filtering the 'delivery_df' DataFrame to only include rows where 'inning' is equal to 2
delivery_df = delivery_df[delivery_df['inning'] == 2]
```

```
In [101]: delivery_df.head()
```

Out[101]:

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	bowler	non_striker	batsman_runs	extra_runs	total_ru
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	AB Dinda	W Jaffer	1	0	
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	AB Dinda	R Dravid	0	1	
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	AB Dinda	R Dravid	0	0	
127	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	4	W Jaffer	AB Dinda	R Dravid	1	0	
128	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	5	R Dravid	AB Dinda	W Jaffer	1	0	

```
In [102]: delivery_df.columns
```

Out[102]: Index(['match_id', 'city', 'winner', 'total_runs_x', 'inning', 'batting_team', 'bowling_team', 'over', 'ball', 'batter', 'bowler', 'non_striker', 'batsman_runs', 'extra_runs', 'total_runs_y', 'extras_type', 'is_wicket', 'player_dismissed', 'dismissal_kind', 'fielder'], dtype='object')


```
In [103]: # Calculating the cumulative sum of 'total_runs_y' for each 'match_id' to create a new column 'current_score'
delivery_df['current_score'] = delivery_df.groupby('match_id').cumsum()['total_runs_y']
```

```
In [104]: delivery_df.head(20)
```

Out[104]:

	match_id	city	winner	total_runs_x	inning	battling_team	bowling_team	over	ball	batter	...	non_striker	batsman_runs	extra_runs	total_runs_y
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	...	W Jaffer	1	0	1
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	...	R Dravid	0	1	1
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	...	R Dravid	0	0	0
127	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	4	W Jaffer	...	R Dravid	1	0	1
128	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	5	R Dravid	...	W Jaffer	1	0	1
129	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	6	W Jaffer	...	R Dravid	0	0	0
130	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	7	W Jaffer	...	R Dravid	0	0	0
131	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	1	1	R Dravid	...	W Jaffer	0	0	0
132	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	1	2	V Kohli	...	W Jaffer	0	0	0
133	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	1	3	V Kohli	...	W Jaffer	0	4	4
134	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	1	4	V Kohli	...	W Jaffer	1	0	1
135	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	1	5	W Jaffer	...	V Kohli	0	0	0
136	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	1	6	W Jaffer	...	V Kohli	0	0	0
137	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	2	1	V Kohli	...	W Jaffer	0	0	0
138	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	2	2	V Kohli	...	W Jaffer	0	0	0

```
In [105]: # Calculating the remaining runs ('runs_left') by subtracting 'current_score' from 'total_runs_x'
delivery_df['runs_left']=delivery_df['total_runs_x']- delivery_df['current_score']
```

```
In [106]: delivery_df.head()
```

```
Out[106]:
```

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	...	batsman_runs	extra_runs	total_runs_y	extras_type
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	...	1	0	1	NaN
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	...	0	1	1	wides
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	...	0	0	0	NaN
127	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	4	W Jaffer	...	1	0	1	NaN
128	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	5	R Dravid	...	1	0	1	NaN

5 rows × 22 columns

```
In [107]: # Calculating the number of balls remaining ('balls_left') by first determining the total balls bowled so far
# (which is calculated by multiplying the number of overs ('over') by 6 and adding the balls bowled in the current over ('ball')).
# Then, subtracting this total from 126, which is the total number of balls available in a 21-over match (6 overs per inning).
# The result is the number of balls left for the team to face in their innings.
delivery_df['balls_left']=126-(delivery_df['over']*6 + delivery_df['ball'])
```

```
In [108]: delivery_df.head()
```

```
Out[108]:
```

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	...	extra_runs	total_runs_y	extras_type	is_wicket	player_dismissed
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	...	0	1	NaN	0	
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	...	1	1	wides	0	
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	...	0	0	NaN	0	
127	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	4	W Jaffer	...	0	1	NaN	0	
128	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	5	R Dravid	...	0	1	NaN	0	

5 rows × 23 columns

```
In [109]: # Replacing any missing (NaN) values in the 'player_dismissed' column with the string "0"
delivery_df['player_dismissed']= delivery_df['player_dismissed'].fillna("0")
```

```
In [110]: delivery_df.head()
```

```
Out[110]:
```

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	...	extra_runs	total_runs_y	extras_type	is_wicket	player_dismissed
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	...	0	1	NaN	0	
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	...	1	1	wides	0	
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	...	0	0	NaN	0	
127	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	4	W Jaffer	...	0	1	NaN	0	
128	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	5	R Dravid	...	0	1	NaN	0	

5 rows × 23 columns

```
In [111]: # Applying a transformation to the 'player_dismissed' column:
# If the value is '0', it remains '0', otherwise, it is changed to '1' (indicating the player was dismissed)
delivery_df['player_dismissed'] = delivery_df['player_dismissed'].apply(lambda x: x if x == '0' else '1')
```

```
In [112]: delivery_df.head()
```

```
Out[112]:
```

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	...	extra_runs	total_runs_y	extras_type	is_wicket	pla
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	...	0	1	NaN	0	
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	...	1	1	wides	0	
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	...	0	0	NaN	0	
127	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	4	W Jaffer	...	0	1	NaN	0	
128	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	5	R Dravid	...	0	1	NaN	0	

5 rows x 23 columns

```
In [113]: # Converting the 'player_dismissed' column to an integer data type
delivery_df['player_dismissed'] = delivery_df['player_dismissed'].astype('int')
```

```
In [114]: wicket = delivery_df.groupby('match_id').cumsum()['player_dismissed'].values
```

```
In [115]: wicket
```

```
Out[115]: array([0, 0, 0, ..., 2, 2, 2])
```

```
In [116]: # Subtracting the 'wicket' values from 10 and storing the result in a new column 'wicket' in the 'delivery_df' DataFrame
delivery_df['wicket'] = 10 - wicket
```

```
In [117]: #Current Run Rate
delivery_df['crr'] = (delivery_df['current_score']*6)/(120-delivery_df['balls_left'])
```

```
In [118]: delivery_df.tail()
```

```
Out[118]:
```

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	...	extras_type	is_wicket	player_dismissed	dismissal
148102	1426312	Chennai	Kolkata Knight Riders	113	2	Kolkata Knight Riders	Sunrisers Hyderabad	9	5	SS Iyer	...	NaN	0		0
148103	1426312	Chennai	Kolkata Knight Riders	113	2	Kolkata Knight Riders	Sunrisers Hyderabad	9	6	VR Iyer	...	NaN	0		0
148104	1426312	Chennai	Kolkata Knight Riders	113	2	Kolkata Knight Riders	Sunrisers Hyderabad	10	1	VR Iyer	...	NaN	0		0
148105	1426312	Chennai	Kolkata Knight Riders	113	2	Kolkata Knight Riders	Sunrisers Hyderabad	10	2	SS Iyer	...	NaN	0		0
148106	1426312	Chennai	Kolkata Knight Riders	113	2	Kolkata Knight Riders	Sunrisers Hyderabad	10	3	VR Iyer	...	NaN	0		0

5 rows × 25 columns

```
In [119]: #Required Run Rate
```

```
delivery_df['rrr']=(delivery_df['runs_left']*6)/(delivery_df['balls_left'])
```

```
In [120]: # Function to determine if the batting team won the match
```

```
def result(row):  
    """  
    This function checks if the 'batting_team' is the same as the 'winner'.  
    If the batting team won, it returns 1; otherwise, it returns 0.  
    Returns:  
    int: 1 if the batting team won, 0 otherwise.  
    """  
    return 1 if row['batting_team'] == row['winner'] else 0
```

```
In [121]: # Applying the 'result' function to each row of 'delivery_df'
```

```
# and storing the output in a new column 'result'  
delivery_df['result'] = delivery_df.apply(result, axis=1)
```

```
In [122]: delivery_df.head()
```

```
Out[122]:
```

	match_id	city	winner	total_runs_x	inning	batting_team	bowling_team	over	ball	batter	...	player_dismissed	dismissal_kind	fielder	current_sc
124	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	1	R Dravid	...	0	NaN	NaN	
125	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	2	W Jaffer	...	0	NaN	NaN	
126	335982	Bangalore	Kolkata Knight Riders	222	2	Royal Challengers Bangalore	Kolkata Knight Riders	0	3	W Jaffer	...	0	NaN	NaN	

```
In [123]: # Selecting relevant columns from 'delivery_df' and creating a new DataFrame 'final_df'
final_df= delivery_df[['batting_team', 'bowling_team', 'city','runs_left','balls_left', 'wicket','total_runs_x', 'crr', 'rrr','result']]
```

```
In [124]: final_df
```

```
Out[124]:
```

	batting_team	bowling_team	city	runs_left	balls_left	wicket	total_runs_x	crr	rrr	result
124	Royal Challengers Bangalore	Kolkata Knight Riders	Bangalore	221	125	10	222	-1.200000	10.808000	0
125	Royal Challengers Bangalore	Kolkata Knight Riders	Bangalore	220	124	10	222	-3.000000	10.845161	0
126	Royal Challengers Bangalore	Kolkata Knight Riders	Bangalore	220	123	10	222	-4.000000	10.731707	0
127	Royal Challengers Bangalore	Kolkata Knight Riders	Bangalore	219	122	10	222	-9.000000	10.770492	0
128	Royal Challengers Bangalore	Kolkata Knight Riders	Bangalore	218	121	10	222	-24.000000	10.809917	0
...	...	...	...	...	...	...	...	...	...	...
148102	Kolkata Knight Riders	Sunrisers Hyderabad	Chennai	3	67	8	113	12.452830	0.268657	1
148103	Kolkata Knight Riders	Sunrisers Hyderabad	Chennai	2	66	8	113	12.333333	0.181818	1
148104	Kolkata Knight Riders	Sunrisers Hyderabad	Chennai	1	65	8	113	12.218182	0.092308	1
148105	Kolkata Knight Riders	Sunrisers Hyderabad	Chennai	0	64	8	113	12.107143	0.000000	1
148106	Kolkata Knight Riders	Sunrisers Hyderabad	Chennai	-1	63	8	113	12.000000	-0.095238	1

71516 rows x 10 columns

```
In [125]: # Generating summary statistics for numerical columns in 'final_df'
final_df.describe()
```

```
Out[125]:
```

	runs_left	balls_left	wicket	total_runs_x	crr	rrr	result
count	71516.000000	71516.000000	71516.000000	71516.000000	7.149800e+04	71516.000000	71516.000000
mean	93.233934	68.789614	7.553051	166.854133	inf	8.517552	0.521240
std	50.325945	33.427842	2.165351	29.843143	NaN	4.502408	0.499552
min	-16.000000	4.000000	0.000000	62.000000	-1.320000e+02	-9.000000	0.000000
25%	54.000000	41.000000	6.000000	147.000000	7.200000e+00	6.428571	0.000000
50%	92.000000	70.000000	8.000000	166.000000	8.571429e+00	8.082139	1.000000
75%	131.000000	98.000000	9.000000	187.000000	1.027397e+01	9.826087	1.000000
max	277.000000	125.000000	10.000000	277.000000	inf	102.000000	1.000000

```
In [126]: # Randomly shuffling all rows of 'final_df' to ensure randomness in the dataset
final_df=final_df.sample(final_df.shape[0])
```

```
In [127]: # Selecting a random row from 'final_df' and displaying it
final_df.sample()
```

```
Out[127]:
```

	batting_team	bowling_team	city	runs_left	balls_left	wicket	total_runs_x	crr	rrr	result
51815	Royal Challengers Bangalore	Kolkata Knight Riders	Bangalore	102	80	9	154	7.8	7.65	1

```
In [128]: # Replacing infinite values with NaN in 'crr' to handle division errors
final_df['crr'].replace([np.inf, -np.inf], np.nan, inplace=True)
```

```
In [129]: # Removing any rows with missing (NaN) values from 'final_df'
final_df.dropna(inplace=True)
```

```
In [130]: # Filtering out rows where 'balls_left' is zero to avoid division errors or irrelevant data
final_df = final_df[final_df['balls_left'] != 0]
```

```
In [131]: # Generating summary statistics for numerical columns in 'final_df'
final_df.describe()
```

```
Out[131]:
```

	runs_left	balls_left	wicket	total_runs_x	crr	rrr	result
count	66815.000000	66815.000000	66815.000000	66815.000000	66815.000000	66815.000000	66815.000000
mean	92.499993	68.37532	7.530001	106.587533	8.884332	8.500164	0.522502
std	50.044087	33.22041	2.180236	29.939326	7.642071	4.488891	0.499497
min	-18.000000	4.000000	0.000000	82.000000	-132.000000	-9.000000	0.000000
25%	53.000000	40.000000	6.000000	147.000000	7.200000	6.428571	0.000000
50%	92.000000	69.000000	8.000000	106.000000	8.554455	8.067227	1.000000
75%	130.000000	97.000000	9.000000	186.000000	10.240000	9.789474	1.000000
max	277.000000	125.000000	10.000000	277.000000	162.000000	102.000000	1.000000

```
In [132]: x = final_df.iloc[:, :-1] # Recreate the features dataframe
y = final_df.iloc[:, -1] # Recreate the target series
```

```
In [133]: print(len(x)) # Should print the number of rows in X (features)
print(len(y)) # Should print the number of rows in y (target)
```

```
66815
66815
```

```
In [134]: # Splitting the dataset into training and testing sets
# 80% of the data will be used for training (x_train, y_train)
# 20% will be used for testing (x_test, y_test)
# random_state=42 ensures reproducibility of the split
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
```

```
In [135]: x_train
```

```
Out[135]:
```

	batting_team	bowling_team	city	runs_left	balls_left	wicket	total_runs_x	crr	rrr
88264	Mumbai Indians	Kolkata Knight Riders	Mumbai	173	121	10	178	-30.000000	8.578512
89305	Mumbai Indians	Royal Challengers Bangalore	Bangalore	13	20	4	142	7.740000	3.900000
85913	Kings XI Punjab	Royal Challengers Bangalore	Chandigarh	130	95	10	175	10.800000	8.210528
26637	Kings XI Punjab	Kolkata Knight Riders	Kolkata	118	82	9	200	12.947368	8.634146
104181	Kolkata Knight Riders	Sunrisers Hyderabad	Kolkata	165	107	9	181	7.384615	9.252336
...	...	...	...	...	...	...	...	...	...
43727	Kings XI Punjab	Mumbai Indians	Mumbai	125	99	10	163	10.857143	7.575758
13726	Rajasthan Royals	Chennai Super Kings	Centurion	149	106	9	164	6.428571	8.433962
131746	Sunrisers Hyderabad	Mumbai Indians	Abu Dhabi	179	98	10	235	15.272727	10.959184
129162	Chennai Super Kings	Royal Challengers Bangalore	Sharjah	137	112	10	156	14.250000	7.339286
37975	Royal Challengers Bangalore	Kings XI Punjab	Dharamsala	164	67	6	232	7.698113	14.686567

53452 rows x 9 columns

```
In [136]: # Display a summary of the x_train DataFrame, including the number of rows, columns,
# data types, non-null values in each column, and memory usage.
x_train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 53452 entries, 88264 to 37975
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   batting_team 53452 non-null  object
1   bowling_team 53452 non-null  object
2   city         53452 non-null  object
3   runs_left    53452 non-null  int64
4   balls_left   53452 non-null  int64
5   wicket       53452 non-null  int32
6   total_runs_x 53452 non-null  int64
7   crr          53452 non-null  float64
8   rrr          53452 non-null  float64
dtypes: float64(2), int32(1), int64(3), object(3)
memory usage: 3.9+ MB
```

```
In [137]: # Import ColumnTransformer to apply different preprocessing steps to specific columns.
# Import OneHotEncoder to convert categorical variables into numerical format using one-hot encoding.
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
```



```

In [138]: # Define a ColumnTransformer to apply OneHotEncoding to categorical columns.
# - 'batting_team', 'bowling_team', and 'city' are transformed using OneHotEncoder.
# - 'sparse=False' ensures the output is a dense NumPy array instead of a sparse matrix.
# - 'drop='first'' removes the first category in each column to avoid multicollinearity.
# - 'remainder='passthrough'' keeps the remaining columns unchanged.
trf = ColumnTransformer([
    ('trf', OneHotEncoder(sparse=False, drop='first')), ['batting_team', 'bowling_team', 'city'])
], remainder='passthrough')

In [139]: # Import LogisticRegression, a linear model used for classification tasks.
from sklearn.linear_model import LogisticRegression

# Import Pipeline to chain multiple preprocessing steps and a model into a single workflow.
from sklearn.pipeline import Pipeline

In [140]: # Create a machine Learning pipeline that automates preprocessing and model training.
# - 'step1': Apply the ColumnTransformer ('trf') to preprocess categorical features using OneHotEncoding.
# - 'step2': Train a Logistic Regression model with the 'Liblinear' solver,
#           which is efficient for small datasets and binary classification tasks.
pipe=Pipeline(steps=[
    ('step1',trf),
    ('step2', LogisticRegression(solver='liblinear'))
])

In [141]: # Train the pipeline on the training data.
# - The ColumnTransformer ('step1') applies OneHotEncoding to categorical features.
# - The LogisticRegression model ('step2') learns patterns from the transformed data
pipe.fit(x_train, y_train)

Out[141]: Pipeline(steps=[('step1',
                           ColumnTransformer(remainder='passthrough',
                                                transformers=[('trf',
                                                                OneHotEncoder(drop='first',
                                                                sparse=False),
                                                                ['batting_team',
                                                                'bowling_team', 'city'])])),
                           ('step2', LogisticRegression(solver='liblinear')))]

In [142]: # Use the trained pipeline to make predictions on the test data.
# - The pipeline first applies the necessary preprocessing (OneHotEncoding).
# - Then, the trained Logistic Regression model predicts the target values.
y_pred = pipe.predict(x_test)

In [143]: y_pred
Out[143]: array([0, 0, 1, ..., 1, 1, 0], dtype=int64)

In [144]: from sklearn.metrics import accuracy_score
accuracy_score(y_test, y_pred )

Out[144]: 0.806106413230562

```

```

In [145]: #RandomForest
from sklearn.ensemble import RandomForestClassifier

In [146]: pipe=Pipeline(steps=[
    ('step1',trf),
    ('step2', RandomForestClassifier())
])

In [147]: pipe.fit(x_train, y_train)
Out[147]: Pipeline(steps=[('step1',
    ColumnTransformer(remainder='passthrough',
        transformers=[('trf',
            OneHotEncoder(drop='first',
                sparse=False),
            ['batting_team',
                'bowling_team', 'city'])])),
    ('step2', RandomForestClassifier())])

In [148]: from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test, y_pred )
print(f"RandomForestClassifier: {accuracy}")
RandomForestClassifier: 0.806106413230562

In [149]: # Import DecisionTreeClassifier
from sklearn.tree import DecisionTreeClassifier

In [150]: # Create a pipeline with ColumnTransformer and DecisionTreeClassifier
pipe = Pipeline(steps=[
    ('step1', trf), # Using the previously defined ColumnTransformer
    ('step2', DecisionTreeClassifier()) # Decision Tree Classifier
])

In [151]: # Fit the pipeline to the training data
pipe.fit(x_train, y_train)
Out[151]: Pipeline(steps=[('step1',
    ColumnTransformer(remainder='passthrough',
        transformers=[('trf',
            OneHotEncoder(drop='first',
                sparse=False),
            ['batting_team',
                'bowling_team', 'city'])])),
    ('step2', DecisionTreeClassifier())])

In [152]: # Make predictions on the test set
y_pred = pipe.predict(x_test)

In [153]: # Calculate accuracy
from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test, y_pred)
print(f"Decision Tree Accuracy: {accuracy}")
Decision Tree Accuracy: 0.9927411509391604

In [154]: import pickle
pickle.dump(pipe, open('model.pkl', 'wb'))

```

```

You, 8 minutes ago | 1 author (You)
1 import streamlit as st
2 import pandas as pd
3 import pickle
4 import plotly.graph_objs as go
5 import plotly.express as px
6 import numpy as np
7 from PIL import Image
8 import base64
9 import matplotlib.pyplot as plt
10 import io
11
12 # Set page configuration
13 st.set_page_config(
14     page_title="IPL Match Predictor",
15     page_icon="🏏",
16     layout="wide"
17 )
18
19 # Custom CSS for styling
20 st.markdown("""
21 <style>
22 .title {
23     color: #2c3e50;
24     text-align: center;
25     font-size: 3em;
26     font-weight: bold;
27     margin-bottom: 30px;
28     background: linear-gradient(45deg, #ff4757, #3742fa);
29     -webkit-background-clip: text;
30     -webkit-text-fill-color: transparent;
31     text-shadow: 1px 1px 2px rgba(0,0,0,0.1);
32 }
33
34 .subtitle {
35     color: #2c3e50;
36     text-align: center;
37     font-size: 1.5em;
38     margin-bottom: 20px;
39 }
40
41 .prediction-container {
42     background-color: #f8f9fa;
43     border-radius: 15px;
44     padding: 25px;
45     box-shadow: 0 6px 12px rgba(0,0,0,0.15);
46     margin-bottom: 30px;
47 }
48
49 .team-prediction {
50     display: flex;
51     justify-content: space-between;
52     align-items: center;
53     margin-bottom: 20px;
54     padding: 10px;
55     background-color: white;
56     border-radius: 10px;
57     box-shadow: 0 2px 6px rgba(0,0,0,0.1);
58 }

```

```

60 .team-name {
61   font-weight: bold;
62   font-size: 1.4em;
63   display: flex;
64   align-items: center;
65   color: #333;
66 }
67
68 .probability-bar {
69   width: 100%;
70   background-color: #e0e0e0;
71   border-radius: 10px;
72   overflow: hidden;
73   margin: 0 20px;
74   height: 25px;
75 }
76
77 .probability-value {
78   font-weight: bold;
79   font-size: 1.3em;
80   color: #333;
81 }
82
83 .icon {
84   font-size: 2.2em;
85   margin-right: 10px;
86 }
87
88 .analysis-container {
89   background-color: #fff;
90   border-radius: 15px;
91   padding: 20px;
92   box-shadow: 0 4px 8px rgba(0,0,0,0.1);
93   margin-bottom: 20px;
94 }
95
96 .analysis-container {
97   background-color: #fff;
98   border-radius: 15px;
99   padding: 20px;
100  box-shadow: 0 4px 8px rgba(0,0,0,0.1);
101  margin-bottom: 20px;
102 }
103
104 .analysis-title {
105   font-size: 1.8em;
106   font-weight: bold;
107   margin-bottom: 15px;
108   color: #2c3e50;
109   border-bottom: 2px solid #2c3e50;
110   padding-bottom: 10px;
111 }
112
113 .analysis-subtitle {
114   font-size: 1.4em;
115   font-weight: bold;
116   margin: 15px 0;
117   color: #3498db;
118 }
119
120 .factor {
121   background-color: #f1f9ff;
122   border-left: 4px solid #3498db;
123   padding: 10px 15px;
124   margin-bottom: 10px;
125   border-radius: 5px;
126 }
127
128 .positive-factor {
129   background-color: #e8f5e9;
130   border-left: 4px solid #4caf50;

```

```

115     padding: 10px 15px;
116     margin-bottom: 10px;
117     border-radius: 5px;
118 }
119
120 .positive-factor {
121     background-color: #e8f5e9;
122     border-left: 4px solid #4caf50;
123 }
124
125 .negative-factor {
126     background-color: #ffebee;
127     border-left: 4px solid #f44336;
128 }
129
130 .neutral-factor {
131     background-color: #fff8e1;
132     border-left: 4px solid #ffc107;
133 }
134
135 .match-situation {
136     display: flex;
137     justify-content: space-around;
138     flex-wrap: wrap;
139     margin: 20px 0;
140 }
141
142 .situation-card {
143     background-color: white;
144     padding: 15px;
145     border-radius: 10px;
146     margin: 10px;
147     box-shadow: 0 2px 5px rgba(0,0,0,0.1);
148     text-align: center;
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
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196
197
198
199
200
201
202
203
204 # Team and city lists
205 teams = ['Sunrisers Hyderabad',
206 'Mumbai Indians',
207 'Royal Challengers Bangalore',
208 'Kolkata Knight Riders',
209 'Kings XI Punjab',
210 'Chennai Super Kings',
211 'Rajasthan Royals',
212 'Delhi Capitals']
213
214 cities = ['Hyderabad', 'Bangalore', 'Mumbai', 'Indore', 'Kolkata', 'Delhi',
215 'Chandigarh', 'Jaipur', 'Chennai', 'Cape Town', 'Port Elizabeth',
216 'Durban', 'Centurion', 'East London', 'Johannesburg', 'Kimberley',
217 'Bloemfontein', 'Ahmedabad', 'Cuttack', 'Nagpur', 'Dharamsala',
218 'Visakhapatnam', 'Pune', 'Raipur', 'Ranchi', 'Abu Dhabi',
219 'Sharjah', 'Mohali', 'Bengaluru']
220
221 # Mapping team names to emoji icons and colors
222 team_data = {
223     'Mumbai Indians': {'emoji': '👑', 'color': '#004BA0', 'secondary_color': '#B7D4ED'},
224     'Chennai Super Kings': {'emoji': '👑', 'color': '#FFFF00', 'secondary_color': '#F9F7A6'},
225     'Royal Challengers Bangalore': {'emoji': '👑', 'color': '#EC1C24', 'secondary_color': '#F7A7A9'},
226     'Kolkata Knight Riders': {'emoji': '👑', 'color': '#3A225D', 'secondary_color': '#B9A6CC'},
227     'Sunrisers Hyderabad': {'emoji': '👑', 'color': '#FF822A', 'secondary_color': '#FFCCA1'},
228     'Delhi Capitals': {'emoji': '👑', 'color': '#00008B', 'secondary_color': '#A6A6ED'},
229     'Kings XI Punjab': {'emoji': '👑', 'color': '#A71930', 'secondary_color': '#E9A8B1'},
230     'Rajasthan Royals': {'emoji': '👑', 'color': '#2F9BE3', 'secondary_color': '#AFDCF8'}
231 }
232
233 # Function to create gauge chart for win probability
234 def create_gauge_chart(win_prob, team_name):
235     team_color = team_data.get(team_name, {}).get('color', '#3498db')

```

```

# Function to create gauge chart for win probability
def create_gauge_chart(win_prob, team_name):
    team_color = team_data.get(team_name, {}).get('color', '#3498db')

    fig = go.Figure(go.Indicator(
        mode = "gauge+number",
        value = win_prob * 100,
        domain = {'x': [0, 1], 'y': [0, 1]},
        title = {'text': f"{team_name} Win Probability", 'font': {'size': 24}},
        gauge = {
            'axis': {'range': [None, 100], 'tickwidth': 1, 'tickcolor': "darkblue"},
            'bar': {'color': team_color},
            'bgcolor': "white",
            'borderwidth': 2,
            'bordercolor': "gray",
            'steps': [
                {'range': [0, 20], 'color': '#ffebee'},
                {'range': [20, 40], 'color': '#ffcdd2'},
                {'range': [40, 60], 'color': '#fff9c4'},
                {'range': [60, 80], 'color': '#c8e6c9'},
                {'range': [80, 100], 'color': '#a5d6a7'}
            ],
            'threshold': {
                'line': {'color': "red", 'width': 4},
                'thickness': 0.75,
                'value': 50
            }
        }
    ))

    fig.update_layout(
        height=300,
        margin=dict(l=20, r=20, t=50, b=20),
        paper_bgcolor="white",

```

```

263     fig.update_layout(
264         height=300,
265         margin=dict(l=20, r=20, t=50, b=20),
266         paper_bgcolor="white",
267         font={"color": "#2c3e50", "family": "Arial"}
268     )
269
270     return fig
271
272 # Function to create run chase visualization
273 def create_runrate_chart(crr, rrr):
274     fig = go.Figure()
275
276     fig.add_trace(go.Bar(
277         x=['Current Run Rate', 'Required Run Rate'],
278         y=[crr, rrr],
279         text=[f'{crr:.2f}', f'{rrr:.2f}'],
280         textposition='auto',
281         marker_color=['#3498db', '#e74c3c'] if crr < rrr else ['#2ecc71', '#e74c3c'],
282     ))
283
284     fig.update_layout(
285         title='Run Rate Comparison',
286         xaxis_title='Run Rate Type',
287         yaxis_title='Runs per Over',
288         height=300,
289         template='plotly_white',
290         margin=dict(l=20, r=20, t=50, b=20)
291     )
292
293     return fig
294

```

```

# Function to analyze match situation
def analyze_match_situation(batting_team, bowling_team, runs_left, balls_left, wickets_left, crr, rrr, target):
    factors = []
    batting_factors = []
    bowling_factors = []

    # Run rate analysis
    if crr > rrr + 2:
        batting_factors.append(f"Current run rate ({crr:.2f}) is significantly higher than required ({rrr:.2f}) - a very comfortable position")
    elif crr > rrr:
        batting_factors.append(f"Current run rate ({crr:.2f}) is higher than required ({rrr:.2f}) - a good position")
    elif rrr > crr + 4:
        bowling_factors.append(f"Required run rate ({rrr:.2f}) is much higher than current run rate ({crr:.2f}) - very challenging chase")
    elif rrr > crr + 2:
        bowling_factors.append(f"Required run rate ({rrr:.2f}) is substantially higher than current run rate ({crr:.2f}) - difficult chase")
    elif rrr > crr:
        factors.append(f"Required run rate ({rrr:.2f}) is slightly higher than current run rate ({crr:.2f}) - pressure on batting side")

    # Wickets analysis
    if wickets_left >= 8:
        batting_factors.append(f"With {wickets_left} wickets in hand, {batting_team} has plenty of batting resources")
    elif wickets_left >= 6:
        batting_factors.append(f"With {wickets_left} wickets remaining, {batting_team} has good batting depth")
    elif wickets_left <= 2:
        bowling_factors.append(f"With only {wickets_left} wickets left, {batting_team} is under significant pressure")
    elif wickets_left <= 4:
        bowling_factors.append(f"With just {wickets_left} wickets remaining, {batting_team}'s batting is limited")
    else:
        factors.append(f"{batting_team} has {wickets_left} wickets remaining")

    # Runs vs balls analysis
    if runs_left > balls_left * 2:
        bowling_factors.append(f"Needing {runs_left} runs from {balls_left} balls (over 12 runs per over) is extremely challenging")
    elif runs_left > balls_left * 1.5:
        bowling_factors.append(f"Needing {runs_left} runs from {balls_left} balls (over 9 runs per over) presents a significant challenge")
    elif runs_left > balls_left:

def create_progression_chart(score, target, overs, wicket):
    # Create data for expected score progression
    ideal_progression = [target * i/20 for i in range(21)]
    current_progression = [0] * 21
    over_int = int(overs)
    over_decimal = overs - over_int

    for i in range(over_int + 1):
        if i < over_int:
            current_progression[i] = score * i / overs
        else:
            current_progression[i] = score

    # Create figure
    fig = go.Figure()

    # Add traces
    fig.add_trace(go.Scatter(
        x=list(range(21)),
        y=ideal_progression,
        mode='lines',
        name='Target Progression',
        line=dict(color='#e74c3c', dash='dash')
    ))

    fig.add_trace(go.Scatter(
        x=list(range(over_int + 1)),
        y=current_progression[:over_int + 1],
        mode='lines+markers',
        name='Current Progression',
        line=dict(color='#3498db'),
        marker=dict(size=8)
    ))

    # Add wickets as markers

```



```

403 # Add wickets as markers
404 if wicket > 0:
405     wicket_positions = np.linspace(1, overs, wicket)
406     wicket_scores = [score * pos / overs for pos in wicket_positions]
407
408     fig.add_trace(go.Scatter(
409         x=wicket_positions,
410         y=wicket_scores,
411         mode='markers',
412         name='Wickets',
413         marker=dict(
414             symbol='x',
415             size=12,
416             color='#e74c3c',
417             line=dict(width=2)
418         )
419     ))
420
421 # Update layout
422 fig.update_layout(
423     title='Match Progression',
424     xaxis_title='Overs',
425     yaxis_title='Runs',
426     height=350,
427     template='plotly_white',
428     xaxis=dict(
429         tickmode='linear',
430         tick0=0,
431         dtick=2
432     ),
433     yaxis=dict(
434         range=[0, target * 1.1]
435     ),
436     legend=dict(
437         orientation="h",
438         vanchor="bottom",
439     )
440 )
441
442 def create_wicket_resources_chart(wickets_left):
443     # T20 batting resources percentages (approximate)
444     resources = [0, 5.5, 13, 22, 32, 43, 55, 67, 80, 92, 100]
445
446     fig = go.Figure(go.Indicator(
447         mode = "gauge+number+delta",
448         value = resources[wickets_left],
449         delta = {'reference': 50},
450         domain = {'x': [0, 1], 'y': [0, 1]},
451         title = {'text': "Batting Resources Remaining", 'font': {'size': 18}},
452         gauge = {
453             'axis': {'range': [None, 100], 'tickwidth': 1, 'tickcolor': "darkblue"},
454             'bar': {'color': "#2ecc71" if resources[wickets_left] > 50 else "#e67e22" if resources[wickets_left] > 25 else "#e74c3c"},
455             'bgcolor': "white",
456             'borderwidth': 2,
457             'bordercolor': "gray",
458             'steps': [
459                 {'range': [0, 25], 'color': '#ffccdd'},
460                 {'range': [25, 50], 'color': '#ffecb3'},
461                 {'range': [50, 75], 'color': '#c8e6c9'},
462                 {'range': [75, 100], 'color': '#a5d6a7'}
463             ],
464         }
465     ))
466
467     fig.update_layout(
468         height=250,
469         margin=dict(l=20, r=20, t=30, b=20),
470     )
471
472     return fig

```

Ethical Form:

Risk Research Ethics Approval

Project Information

Project Ref	7
Full Name	Nitesh Karki
Faculty	Faculty of
Department	School of
Supervisor	Manoj Shrestha
Module Code	ST6047CEM
EFAAF Number	EFAAF
Project Title	Research and Application of Machine Learning Techniques to Determine Indian Premier League (IPL) Winner Prediction
Date(s)	Date(s)
Created	Created

Project Summary

This research aims to apply machine learning techniques to predict the winner of IPL matches by analyzing historical data, player performance, and team statistics. The study will explore algorithms like decision trees, random forests, and neural networks to identify key factors influencing match outcomes. Additionally, it will assess the impact of external elements such as weather conditions and player injuries on prediction accuracy. The goal is to develop a model that can accurately forecast IPL match results and provide valuable insights for teams and analysts.

Names of Co-Investigators and their organisational affiliation(place of study /employer)	
Is this project externally funded?	No
Are you required to use a Professional Code of Ethical Practice appropriate to your discipline?	Yes
Have you read the Code?	Yes

Project Details

What are the aims and objectives of the project?	The goal of this study is to utilize artificial intelligence and machine learning to predict IPL match outcomes. The project will analyze past IPL match data (2008-2017) to identify patterns that influence results, focusing on team and player dynamics. Machine learning models, such as classification algorithms, will be developed and trained to forecast match winners, with performance evaluated using accuracy, precision, and recall metrics. The impact of external factors like weather, player injuries, and team strategies will also be considered to enhance prediction accuracy. Ultimately, the study aims to provide actionable insights for IPL teams, analysts, and fans, demonstrating the potential of machine learning in sports analytics.
Explain your research design	The research design for this project involves using machine learning to predict IPL match outcomes. First, historical match data from 2008 to 2017 will be collected, including player and team statistics, along with external factors like weather and injuries. This data will be preprocessed and analyzed to identify key features influencing match results. Machine learning models, such as decision trees and random forests, will be trained to predict match winners, with their performance evaluated using metrics like accuracy and precision. External factors will be integrated into the models to improve prediction accuracy. The final model will provide actionable insights for IPL teams, analysts, and fans, demonstrating the potential of machine learning in sports analytics.
Outline the principal methods you will use	Data Collection: Gather historical IPL match data (2008-2017), including match-by-match statistics, player performance metrics, and team statistics. External factors such as weather, injuries, and team strategies will also be collected.

ed based on their potential influence on match outcomes. Feature Selection and Engineering: Identify and create relevant features, such as player form, team strength, and venue conditions, that could improve model accuracy. This step involves both feature selection (choosing the most important factors) and feature engineering (creating new variables from existing data). Machine Learning Models: Develop and train various machine learning classification algorithms, such as decision trees, random forests, and support vector machines, to predict the outcome of matches. These models will be trained on the selected features. Model Evaluation: Evaluate the performance of the models using accuracy, precision, recall, and other metrics. Cross-validation will be used to ensure the models are robust and not overfitting. External Factors Integration: Incorporate external factors like weather conditions, player injuries, and team strategies into the models to assess their impact on match predictions and improve accuracy. Prediction and Insight Generation: Use the final model to predict future IPL match outcomes and generate actionable insights for teams, analysts, and fans on how machine learning can enhance sports analytics.	
Are you proposing to use a validated scale or published research method / tool?	Yes
Does your research seek to understand, identify, analyse and/or report on information on terrorism or from terrorist organisations, require access to terrorist groups or those convicted of terrorist offences or relate to terrorism policies in other international jurisdictions?	No
Does your research seek to understand, identify, analyse and/or report on information for other activities considered illegal in the UK and/or in the country you are researching in?	No
Are you dealing with Secondary Data? (e.g. sourcing info from websites, historical documents)	Yes

Are you dealing with personal data?	No
Please specify what personal data you will be collecting.	
Are you dealing with sensitive data (special category data)?	No
Will the Personal or Sensitive data be shared with a third party?	No
Will the Personal or Sensitive data be shared outside of the European Economic Area (EEA)?	No
Is the project solely desk based? (e.g. involving no laboratory, workshop or offcampus work or other activities which pose significant risks to researchers or participants)	Yes
Will the data collection, recruitment materials or any other project documents be in any language other than English?	Yes
Are there any other ethical issues or risks of harm raised by the study that have not been covered by previous questions?	No

External Ethics Review

Question	Yes	No
Will this study be submitted for ethical review to an external organisation ? (e.g. Another University, Social Care, National Health Service, Ministry of Defence, Police Service and Probation Office)		×
If YES, name of external organisation		
Will this study be reviewed using the IRAS system?		×
Has this study previously been reviewed by an external organisation?	×	×

DBS (Disclosure & Barring Service) formerly CRB (Criminal Records Bureau)

Question	Yes	No
Does the study require DBS (Disclosure & Barring Service) checks?		×
If YES, Please give details of the level of check, serial number, date obtained and expiry date (if applicable)		
If NO, does the study involve direct contact by any member of the research team with children or young people under 18 years of age?		×
If NO, does the study involve direct contact by any member of the research team with adults who have learning difficulties, brain injury, dementia, degenerative neurological disorders?	×	×
If NO, does the study involve direct contact by any member of the research team with adults who are frail or physically disabled?		×
If NO, does the study involve direct contact by any member of the research team with adults who are living in residential care, social care, nursing homes, rehabilitation centres, hospitals or hospices ?		×
If NO, does the study involve direct contact by any member of the research team with adults who are in prison, remanded on bail or in custody?		×
If you have answered YES to any of the questions above please explain the nature of that contact and what you will be doing		

Confidentiality, security and retention of research data

Question	Yes	No
What data are you collecting / using / recording?		
Are there any reasons why you cannot guarantee the full security and confidentiality of any personal or confidential data collected for the study?		×
Please provide an explanation		
Is there a significant possibility that any of your participants, and associated persons, could be directly or indirectly identified in the outputs or findings from this study?		×
Please provide an explanation		
Is there a significant possibility that a specific organisation or agency or participants could have confidential information identified, as a result of the way you write up the results of the study?		×
Please provide an explanation		
Will any members of the research team retain any personal or confidential data at the end of the project, other than in fully anonymised form?		×
Please provide an explanation		
Will you or any member of the team intend to make use of any confidential information, knowledge, trade secrets obtained for any other purpose than the research project ?		×
Please give an explanation		
Have you taken necessary precautions for secure data management, in accordance with data protection and CU Policy	×	
Specify location (physical and electronic) where data will be stored		
Will you be responsible for destroying the data after study completion?	×	
If NO, who will be responsible for this?		
Please explain how any identifiable and anonymous data will be destroyed	After the study, all data will be securely destroyed to ensure privacy and comply with ethical guidelines. Identifiable data will be deleted using secure software, and backups will be permanently removed.	
Planned disposal date	Within 3 months after the study is completed and results are published.	

Participant Information and Informed Consent

Question	Yes	No
Will all the participants be fully informed BEFORE the project begins why the study is being conducted and what their participation will involve ?		✗
Please explain why		
Will every participant be asked to give written consent to participating in the study, before it begins ?		✗
If NO, please explain how you will get consent from your participants.If not written consent, explain how you will record consent	This study does not involve human participants	
Will all participants be fully informed about what data will be collected, and what will be done with this data during and after the study ?		✗
If NO, please specify	This study does not involve human participants or the collection of personal data. It uses publicly available historical match data and player statistics for analysis.	
Please explain what recordings (audio, visual or both) will be made and how you will gain consent for recording participants		
Will all participants understand that they have the right not to take part at any time, and/or withdraw themselves and their data from the study if they wish?		✗
If NO, please explain why	This study does not involve human participants. It uses publicly available historical match data and player statistics, and therefore there is no need for participants to be informed of their right to withdraw.	
Will every participant understand that there will be no reasons required or repercussions if they withdraw or remove their data from the study?		✗
If NO, please explain why	This study does not involve human participants. It uses publicly available historical match data and player statistics, so there is no need for participants to be informed about withdrawing or removing their data.	
Does the study involve deceiving, or covert observation of, participants?		✗
Will you debrief them at the earliest possible opportunity?		✗
If NO to debrief them, please explain why this is necessary	This study does not involve human participants, deception, or covert observation, so there is no need for a debriefing.	

Risk of harm, potential harm and disclosure of harm

Question	Yes	No
Is there any significant risk that the study may lead to physical harm to participants or researchers ?		×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case		
Is there any risk that your study may lead or result in harm to the reputation of the University Group, its researchers or the organisations involved in the study?		×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case		
Is there a risk that the study will lead to participants to disclose evidence of previous criminal offences, or their intention to commit criminal offences?		×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case		
Is there a risk that the study will lead participants to disclose evidence that children or vulnerable adults are being harmed, or at risk or harm?		×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case		
Is there a risk that the study will lead participants to disclose evidence of serious risk of other types of harm ?		×
If you have answered Yes, please explain how you will take steps to reduce or address those risks. If you have answered No, explain why you believe this is the case		
Will participants be made aware of the circumstances in which disclosure has implications for confidentiality?		×

Payments to participants

Question	Yes	No
Do you intend to offer participants cash payments or any kind of inducements, or reward for taking part in your study ?		×
If YES, please explain what kind of payment you will be offering(e.g.prize draw or store vouchers)		
Is there any possibility that such payments or inducements will cause participants to consent to risks that they might not otherwise find acceptable ?		×
If YES, please explain)		
Is there any possibility that the prospect of payment or inducements will influence the data provided by participants in any way ?		×
If YES, please explain)		
Will you inform participants that accepting payments or inducements does not affect their right to withdraw from the study at any time ?		×

Capacity to give valid consent

Question	Yes	No
Do you propose to recruit any participants?		X
Do you propose to recruit any participants who are children or young people under 18 years of age?		X
Do you propose to recruit any participants who are adults who have learning difficulties, mental health conditions, brain injury, advanced dementia, degenerative neurological disorders ?		X
Do you propose to recruit any participants who are adults who are physically disabled and cannot provide written and/or verbal consent		X
Do you propose to recruit any participants who are with adults who are living in residential care, social care, nursing homes, reablement centres, hospitals or hospices ?		X
Do you propose to recruit any participants who are with adults who are in prison, remanded on bail or in custody?		X
If you have answered YES to any of the questions above please explain overcome any challenges to gaining valid consent		
Do you propose to recruit any participants with possible communication difficulties, including difficulties arising from limited use of knowledge of the English language ?		X
If YES, please explain how you will overcome any challenges to gaining valid consent		
Do you propose to recruit participants who may not be able to fully understand the nature of the study, the foreseen implications or cannot provide consent?		X
If YES, please explain how you will overcome any challenges to gaining valid consent		

Recruiting Participants

Question	Yes	No
Who are the participants?		
How are participants being recruited? Please provide details on all methods of recruitment you intend to use		
Do you foresee any conflict of interest?		X
Please explain how will this conflict of interest be addressed		

Online and Internet Research

Question	Yes	No
Will any part of your project involve collecting data via the internet or social media?		X
If YES, please explain how you will obtain permission to collect data by these means		
Will this require consent to access?		X
If NO, please explain how you will get permission/ consent' to collect this information?		
Will you be collecting data using an online questionnaire/ survey tool? (e.g. BoS, Filemaker)?		X
If YES, please explain which software and how you are ensuring appropriate data security		
Is there a possibility that the study will encourage children under 18 to access inappropriate websites, or correspond with people who pose risk of harm ?		X
If YES, please explain further		
Will the study incur any other risks that arise specifically from the use of electronic media ?		X
If YES, please explain further		

Information gathered from human participants

Question	Yes	No
Primary		
Does your project involve primary data collection from human participants via questionnaires, focus groups, interviews, psychological tests, photography/videography etc.?		×
If YES, Please detail the information to be collected and methods that will be used.		
Is there the possibility of physical or psychological harm to the researcher(s) or the participants?		×
If YES, please explain the possible harm and action taken to reduce/remove the risk		
Are any specific exclusions needed to prevent possible harm to participants (e.g. excluding people with known mental health problems)?		×
If YES, please explain exclusions needed and how these will be carried out		
Are any of the questionnaires or other tests being used in the research diagnostic for specific clinical conditions?		×
If YES, Please explain how you will take steps to reduce or address these risks		